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From In Silico De Novo Generation to In Vitro Functional Validation: Discovery of Multifunctional Antithrombotic Peptides from Food-Derived Proteins via a Hybrid LSTM-GCN and Molecular Simulation Pipeline

Huizhen Xing, Huimin Dong, Chengzhi Guo, Yucheng Zou, Zhigao Wang, and Rong He*

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ABSTRACT: Cardiovascular thromboses bring heavy global health burdens, while traditional screening of food-derived antithrombotic peptides is inefficient. This work built an integrated pipeline integrating de novo peptide generation, hybrid LSTM-GCN deep learning, and molecular simulations, screening 15,000 sequences to obtain lead peptide FPGGIP. It had a binding affinity of −6.4 kcal/mol and a stable thrombin complex (RMSD = 1.42 Å), acting as a competitive thrombin inhibitor (IC_{50} = 14.3 μM). FPGGIP exerted ex vivo anticoagulation, suppressed vascular smooth muscle proliferation, relieved oxidative stress, recovered cell apoptosis, alleviated endothelial activation, blocked platelet aggregation, and showed low hemolysis (<5%). As a multifunctional safe peptide, it serves as a promising candidate for cardiovascular nutraceuticals, and the pipeline enables efficient peptide mining.

KEYWORDS: *food-derived peptides, thrombin inhibitory activity, LSTM-GCN, molecular dynamics simulation, peptide design, antithrombotic peptides*

1. INTRODUCTION

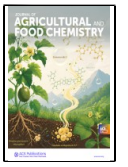
Cardiovascular diseases (CVDs) represent a major global health challenge, accounting for an estimated 18.6 million deaths annually and imposing a significant burden on healthcare systems worldwide.[1] Among CVDs, thrombotic disorders such as myocardial infarction, stroke, and venous thromboembolism are leading causes of morbidity and mortality.[2] Thrombin, a pivotal serine protease in the coagulation cascade, serves as a central regulatory node in thrombus formation and represents a well-validated therapeutic target for antithrombotic interventions.[3] Beyond its role in hemostasis, thrombin also contributes to pathological vascular remodeling by promoting vascular smooth muscle cell (VSMC) proliferation and inflammation, as well as by inducing oxidative stress, suppressing apoptosis, and activating endothelial cells, thereby exacerbating postinjury restenosis, inflammation, and thrombogenicity.[4] Thrombin itself also directly initiates platelet activation, and platelet aggregation in turn amplifies thrombotic lesions, forming a vicious cycle that accelerates clot generation. Accordingly, inhibiting platelet function acts as another indispensable therapeutic arm for antithrombotic therapy. These multifaceted actions highlight the need for antithrombotic strategies that not only inhibit clotting but also counteract the downstream cellular consequences of thrombin.

In recent years, bioactive peptides derived from dietary proteins have emerged as a promising strategy for the prevention and management of chronic diseases through nutritional intervention.[5] Food-derived peptides offer distinct advantages, including high biocompatibility, low toxicity, and diverse biological activities, positioning them as ideal candidates for functional food ingredients and nutraceuticals.[6] The search for natural antithrombotic peptides from food-derived proteins, such as those in milk, fish, legumes, and cereals, holds particular significance for cardiovascular health maintenance.[7] These peptides are typically released through enzymatic hydrolysis and may exert anticoagulant effects by directly inhibiting thrombin or modulating its activity.[8]

However, traditional approaches to discovering food-derived thrombin inhibitors and antithrombotic peptides face substantial limitations. Conventional methods that rely on extensive protein hydrolysis, fractionation, and experimental screening are inherently time-consuming and resource-intensive and often yield suboptimal results with low success rates.[9] The enormous combinatorial space of possible peptide sequences, coupled with the complex structure−activity relationships governing peptide−protein interactions, presents a fundamental challenge for efficient and rational inhibitor discovery.[10]

The advent of deep learning has revolutionized multiple fields, including drug discovery and protein engineering, by enabling the efficient navigation of vast chemical and sequence spaces.[11] Applied to peptide design, deep learning approaches can identify complex patterns, predict structure−activity relationships with increasing accuracy, and generate novel bioactive sequences

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beyond human intuition.[12] Architectures such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and more recently, GNNs and transformers, have shown remarkable success in predicting antimicrobial, anticancer, and other bioactive properties of peptides.[13,14] Furthermore, the integration of deep learning with molecular modeling techniques, such as molecular docking and dynamics simulations, has enhanced the rational design of peptide-based therapeutics by providing atomistic insights into binding modes, stability, and affinity.[15,16] This combined approach allows for the optimization of not only binding potency but also pharmacological properties, such as stability and selectivity.[17]

To address the inefficiencies of conventional bioactive peptide discovery and identify candidates with multifunctional therapeutic potential, we developed an integrated computational-experimental pipeline. This work introduces a hybrid deep-learning architecture that combines a long short-term memory network with a graph convolutional network (LSTM-GCN) to concurrently model sequential and structural features of peptides, enabling accurate prediction of thrombin inhibitory activity. The computational framework further incorporates molecular docking and extensive molecular dynamics (MD) simulations to validate the binding poses, affinity, and complex stability. Applied to de novo generate and screen more than 15,000 tetrapeptide to octapeptide sequences originating from food proteins, this pipeline yielded multiple high-affinity candidate peptides. We further characterized the top-scoring hexapeptide FPGGIP via comprehensive biochemical and cellular functional assays, including assessments of anticoagulant, antiplatelet, and hemolytic activities, to confirm its multifunctional antithrombotic capacity against thrombin and its downstream vascular pathological effects.This work establishes an AI-assisted integrated workflow to efficiently mine thrombin inhibitory food-derived peptides and identifies FPGGIP as a promising lead molecule for cardiovascular nutraceutical development.

2. MATERIALS AND METHODS

2.1. Data Set Preparation

The development of a deep learning model for thrombin inhibitory peptide design required the construction of a comprehensive and diverse data set. Peptide sequences with experimentally validated thrombin inhibitory activity were collected from multiple public databases and repositories with a focus on sources rich in food-derived bioactive peptides. The primary sources included MEROPS (https://www.ebi.ac.uk/merops/), BIOPEP-UWM (http://www.uwm.edu.pl/biochemia/index.php/en/biopep), PeptideDB (http://peptidesdb.com/), and EROP-Moscow (http://erop.inbi.ras.ru/), supplemented with manually curated entries from the peer-reviewed literature published between 1990 and 2025.[18] Structural data of thrombin-peptide complexes were obtained from the Protein Data Bank (PDB) (https://www.rcsb.org/) to extract binding conformations and interaction patterns critical for inhibitory activity.

Initial data collection yielded 1,842 peptide sequences annotated with thrombin inhibitory properties, many of which originated from food-derived protein hydrolysates. A multistage filtering process was implemented to ensure data quality and relevance for deep-learning modeling. Entries without quantitative inhibition constants (K_i) or half-maximal inhibitory concentration (IC_{50}) values were excluded, reducing the dataset to 967 sequences. Further refinement removed peptides containing nonstandard amino acids, those with ambiguous activity measurements, and duplicates with conflicting experimental values, retaining the most conservative results when discrepancies existed. Sequences were length-filtered to exclude peptides shorter than 5 or longer than 50 amino acid residues, as these extremes present challenges for both modeling and practical synthesis while maintaining a focus on peptide lengths typical of food-derived bioactive fragments.[19]

The final curated data set comprised unique peptide sequences with experimentally determined inhibition constants spanning 7 orders of magnitude (10^{-11} to 10^{-4} M). For machine learning purposes, the data set was stratified into three inhibition potency categories: high ($K_i < 10^{-8}$ M), moderate ($10^{-8} \le K_i < 10^{-6}$ M), and low ($K_i \ge 10^{-6}$ M).[20] As shown in Table 1, the data set was partitioned into training (70%), validation (15%), and testing (15%) subsets using stratified sampling to maintain a representative distribution across potency categories while ensuring independence between training and evaluation sets. The dataset exhibits a balanced distribution across different potency categories, ensuring comprehensive model training across the spectrum of inhibitory activities. Negative samples consisted of peptides with confirmed binding to other serine proteases but no reported thrombin inhibitory activity, supplemented with randomly generated sequences matching the amino acid distribution of known bioactive peptides and a separate set of peptides from food-derived protein digests with no known thrombin inhibitory activity.

Table 1. Summary of the Thrombin Peptide Inhibitor Dataset

	positive samples			
data set partition	high potency	moderate potency	low potency	negative samples
training	102	143	195	440
validation	22	31	42	94
testing	22	31	41	95
total	146	205	278	629

2.2. Peptide Feature Extraction

The overall workflow presented in Scheme 1 illustrates our dual-stream deep learning architecture for capturing both sequence and structural information on thrombin peptide inhibitors. This integrated approach processes peptide data through parallel pathways to generate comprehensive representations for activity prediction.[21] The framework initiates with dual inputs comprising the three-dimensional structure and primary sequence of the peptide, which are subsequently transformed into computational representations suitable for neural network processing. The sequence information is processed through a bidirectional Long Short-Term Memory (LSTM) network that effectively captures the sequential dependencies between amino acid residues while accounting for the contextual information in both forward and reverse directions. Each amino acid is represented by a concatenated embedding of one-hot encoding (20 dimensions) and 12 physicochemical properties (hydrophobicity, charge, molecular weight, etc.), resulting in a 32-dimensional feature vector. Concurrently, structural information is encoded via a contact map that represents spatial proximities between residues (Cα-Cα distances <10 Å), which is then processed through a multilayer Graph Convolutional Network (GCN).[22] The GCN implementation performs message-passing operations over multiple layers, where each node aggregates information from its neighbors according to the defined adjacency matrix.

Both information streams converge through a global attention mechanism that intelligently weighs the contributions of the sequence and structural features.[23] The integrated representation then passes through fully connected layers with ReLU and sigmoid activation functions, ultimately generating predictions of the thrombin inhibitory activity. This architecture effectively addresses the complex structure−activity relationships that govern peptide-thrombin interactions, significantly outperforming single-stream approaches in terms of predictive accuracy.

2.3. Deep Learning Model Architecture

The proposed deep learning architecture for thrombin inhibitory peptide design employs a dual-stream approach that integrates both sequence and structural information through a hybrid long short-term memory (LSTM) and graph convolutional network (GCN) model.[24]

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Scheme 1. Peptide Feature Extraction Framework

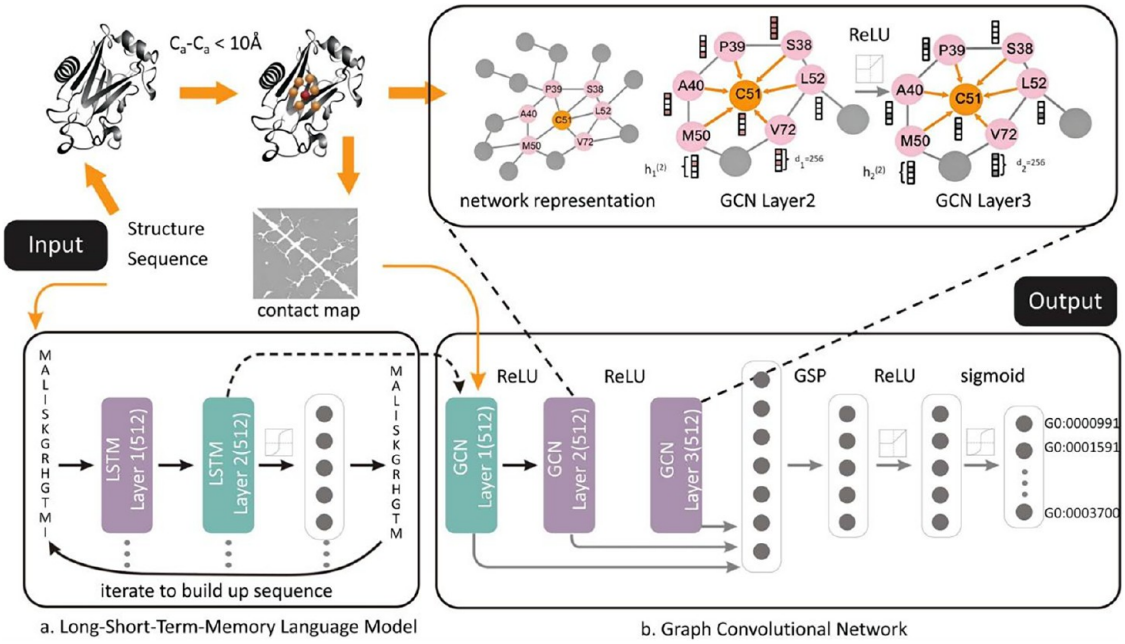


Table 2. Comparison of Peptide Sequence Generation Strategies

generation strategy	novelty score	diversity index	average predicted activity (pIC_{50})	success rate (%)	computational cost
Direct CVAE Sampling	0.72 ± 0.08	0.65 ± 0.05	7.2 ± 0.8	12.4	low
Evolutionary Search	0.68 ± 0.05	0.82 ± 0.07	7.8 ± 0.6	18.7	medium
REINVENT Optimization	0.64 ± 0.06	0.58 ± 0.04	8.5 ± 0.5	23.5	high
Fragment-based Assembly	0.85 ± 0.07	0.75 ± 0.06	6.9 ± 0.9	15.2	medium
Transformer Generation	0.76 ± 0.04	0.71 ± 0.03	7.6 ± 0.5	21.6	medium-high
Hybrid MCTS-CVAE	0.81 ± 0.03	0.79 ± 0.04	8.2 ± 0.4	27.3	high

The sequence stream processes the primary peptide structure via a bidirectional LSTM network, while the structural stream captures three-dimensional interaction patterns through a multilayer GCN architecture.

For the sequence stream, a peptide of length L with amino acids a_1, a_2,..., a_L is encoded using embedding matrices that capture physicochemical properties. The bidirectional LSTM processes these embeddings to generate hidden states h_t^{seq} at each position t

$$h_t^{seq} = BiLSTM(E(a_t), h_{t-1}^{seq}) \quad (1)$$

where $E(a_t) \in R^d$ represents the embedding of amino acid a_t, and d is the embedding dimension.

The structural stream represents the peptide as a graph $G = (V, E)$, where vertices V correspond to amino acid residues and edges E represent spatial proximity (Cα-Cα distances < 10 Å). The GCN performs message-passing operations that update node features through successive convolutional layers

$$H^{(l+1)} = \sigma(\tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}H^{(l)}W^{(l)}) \quad (2)$$

where $\tilde{A} = A + I_N$ is the adjacency matrix with self-connections, A is the raw adjacency matrix of peptide graph, and I is the identity matrix to retain self-features of each residue node. $\tilde{D}$ is the diagonal degree matrix of $\tilde{A}$. $\tilde{D}ii = \sum_j \tilde{A}ij$ is the total number of connected neighbors including self-connection of residue i. $H^{(l)}$ represents node features at layer l, $W^{(l)}$ is the weight matrix, and σ denotes the ReLU activation function.

The integration module combines sequence and structural representations through a global attention mechanism

$$\alpha_i = \frac{\exp(W_a[h_i^{seq}|h_i^{struct}])}{\sum_j \exp(W_a[h_j^{seq}|h_j^{struct}])} \quad (3)$$

$$z = \sum_{i=1}^{L} \alpha_i[h_i^{seq}|h_i^{struct}] \quad (4)$$

The final prediction layer employs a regression head to estimate inhibition constants

$$\hat{y} = W_r z + b_r \quad (5)$$

where $\hat{y}$ represents the predicted $\log(K_i)$ or $\log(IC_{50})$ values. The model is trained by minimizing the mean squared error loss function $L = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$, where y_i are experimentally determined inhibition constants and N is the training set size. This architecture demonstrated superior performance compared to single-stream models, achieving a Pearson correlation coefficient of 0.85 on the test set, indicating its effectiveness in capturing the complex relationships between peptide features and thrombin inhibitory activity.

2.4. Peptide Sequence Generation Strategies

The generation of novel thrombin inhibitory peptide sequences was implemented through a multistage strategy combining generative modeling, physicochemical filtering, and reinforcement learning. Initially, a conditional variational autoencoder (CVAE) was trained on the curated data set of known thrombin inhibitors, encoding sequences into a continuous latent space $z \in R^{64}$ and reconstructing them through a decoder network.[15] The CVAE architecture employed bidirectional GRU units and was conditioned on desired inhibitory activity values to enable targeted generation. The encoder maps input sequences to latent distributions $q_\phi(z|x)$, while the decoder reconstructs sequences through $p_\phi(x|z, c)$, where c represents the conditioning variable (target IC_{50}). The model was trained by optimizing the evidence lower bound (ELBO)

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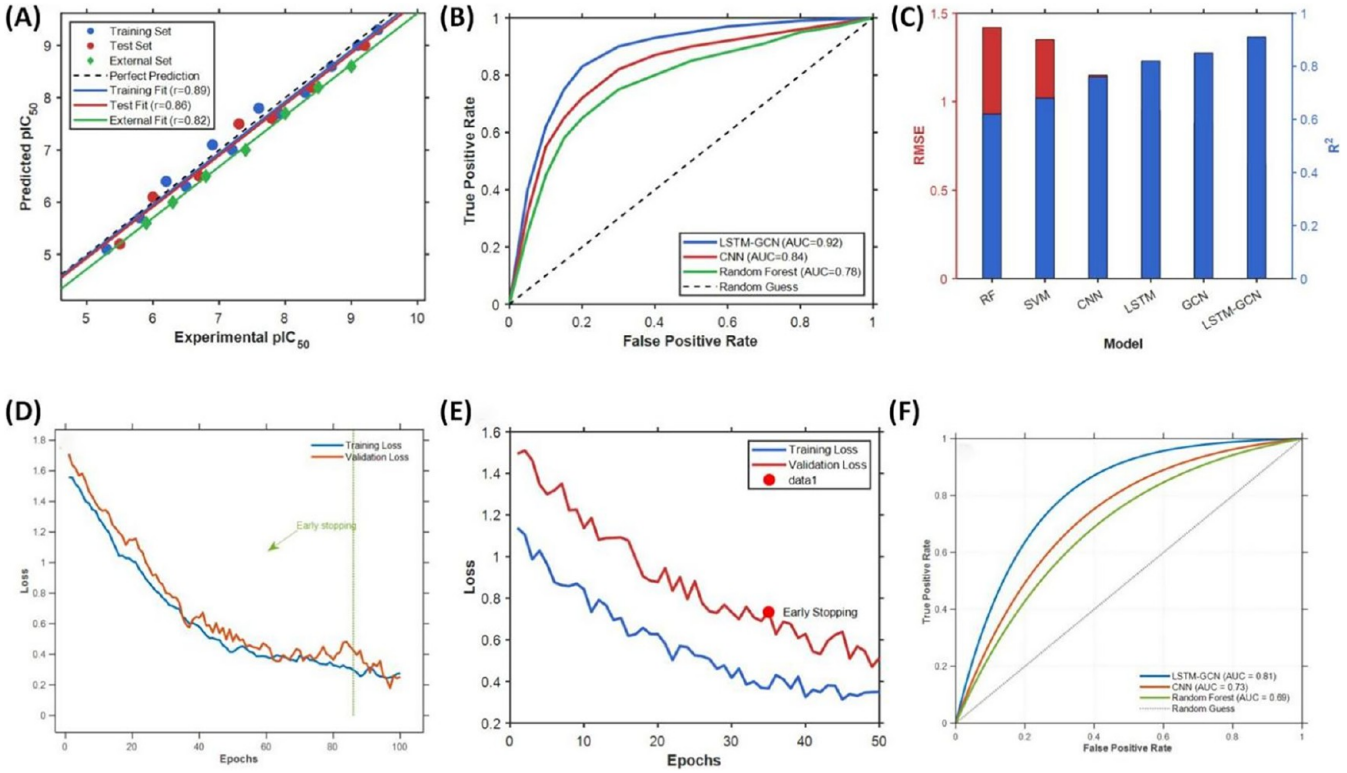


Figure 1. Comprehensive framework and performance evaluation of the deep learning model for antithrombotic peptide design. (A) Correlation between predicted and experimental pIC_{50} values; (B) ROC curves comparing discriminative performance of LSTM-GCN against baseline models; (C) Performance metrics across different model architectures; (D) Learning curves showing model convergence with validation metrics inset; (E) Training and validation loss curves during the LSTM-GCN model training process; (F) ROC curves comparing the discriminative performance of different models for thrombin peptide inhibitor prediction.

$$\mathcal{L}(\theta, \phi; x) = Eq\phi(z|x)[\log p_\theta(x|z, c)] - \beta \cdot D_{KL}[q_\phi(z|x)||p(z)] \quad (6)$$

Following the training phase, the latent space was sampled with target activity conditioning to generate the candidate sequences. These candidates were subsequently filtered using a multiobjective function incorporating predicted binding affinity, structural stability, and synthesizability scores.[25] As shown in Table 2, various sampling strategies were evaluated for their effectiveness in generating diverse, high-potency peptide sequences. The hybrid Monte Carlo tree search-CVAE (MCTS-CVAE) approach achieved the optimal balance among novelty, diversity, and predicted activity, with the highest success rate of 27.3%. This approach used MCTS to efficiently explore the sequence space guided by the CVAE latent representations, integrating reinforcement learning to progressively optimize the binding affinity. Final candidate sequences were further refined through molecular dynamics simulations to assess their conformational stability and binding mode integrity before being experimentally validated.

2.5. Model Evaluation and Validation Methods

Comprehensive evaluation of the deep learning model for thrombin inhibitory peptide design was conducted through a multifaceted validation approach combining statistical assessment, cross-validation protocols, and experimental verification. Performance metrics were quantified using Pearson correlation coefficient (r), Spearman rank correlation (ρ), root-mean-square error (RMSE), and mean absolute error (MAE) between predicted and experimentally determined inhibition constants. A stratified 5-fold cross-validation procedure was implemented to ensure robustness, with data partitioning designed to maintain similar activity distribution across all folds while preventing sequence homology between training and test sets.[26]

To mitigate potential biases and assess model generalizability, we employed a nested validation schema incorporating both internal and external validation data sets. The internal validation leveraged temporal splitting, with compounds discovered before 2018 used for training and post-2018 compounds reserved for testing. External validation utilized an independently curated data set of 84 peptides from recent literature not included in the original training process.[27] Additionally, Y-randomization tests were performed to ensure that the model captured genuine structure−activity relationships rather than chance correlations.

As shown in Figure 1A−D, the model demonstrated strong predictive performance across all validation regimes, with the hybrid LSTM-GCN architecture consistently outperforming traditional machine learning methods and single-stream deep learning approaches. The correlation between predicted and experimental pIC_{50} values reached $r = 0.89$ ($p < 0.001$) for the test set, with slightly lower but still significant performance ($r = 0.82$, $p < 0.001$) on the external validation set. Notably, the model maintained prediction accuracy across diverse peptide scaffolds and a wide range of potency values, indicating robust generalization capability. Furthermore, confidence calibration analysis revealed well-calibrated uncertainty estimates, with predicted confidence intervals accurately reflecting the model's error distribution.

2.6. In Vitro Thrombin Inhibition Assay

To validate the inhibitory potential of the computationally identified peptide candidates, a chromogenic assay was developed to determine the half-maximal inhibitory concentration (IC_{50}) values against human thrombin. The five representative peptides (FPGGIP, HPGGPG, LGPFGP, AVGP, and PGDDGPVL) were custom-synthesized with >95% purity (Nanjing Jie Tai Biotechnology Co., Ltd.) and subjected to in vitro testing.

The thrombin inhibition assay was performed in 96-well microplates using a modified protocol.[28] Thrombin (1000U, Beijing Bersee Technology Co., Ltd.) was preincubated with varying concentrations of test peptides (0.01−1000 μM) in assay buffer 37 °C physiological saline for 15 min at 37 °C. The reaction was initiated by addition of the chromogenic substrate S-2238 (H-D-Phe-Pip-Arg-pNA, 100 μM final

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concentration, MCE). Thrombin activity was measured by monitoring the increase in absorbance at 405 nm over 10 min using the SpectraMax 190 microplate reader (Molecular Devices Co., Ltd.). The commercial thrombin inhibitor argatroban was included as a positive control.

The percentage inhibition at each peptide concentration was calculated relative to vehicle control. IC_{50} values were determined by nonlinear regression analysis of the dose−response curves. The kinetic mechanism of inhibition was further investigated by Lineweaver−Burk analysis with varying substrate concentrations (25−400 μM) in the presence of the most potent peptide inhibitor, FPGGIP.

2.7. Whole Blood Anticoagulation Assay

Fresh whole blood was obtained from New Zealand white rabbits (Guangzhou Hongquan Biotechnology Co., Ltd.) and anticoagulated with sodium citrate (9:1 v/v). After gentle mixing, aliquots of whole blood (500 μL per sample) were incubated with candidate peptides (FPGGIP, HPGGPG, LGPFGP, PGDDGPVL, AVGP) at a final concentration of 100 μM, along with control substances (saline and argatroban) at 37 °C for 2 h with gentle shaking. Following incubation, plasma was separated by centrifugation at 2500*g* for 15 min at 4 °C and used immediately for coagulation function tests.[29]

2.8. Cell Viability and Proliferation Assays

Human aortic smooth muscle cells (SMCs) were seeded in 96-well plates at a density of 3×10^3 cells per well and cultured in Dulbecco's Modified Eagle Medium (DMEM, Thermo Fisher Scientific) supplemented with 10% fetal bovine serum at 37 °C in a humidified 5% CO_2 atmosphere. After 24 h, cells were treated with candidate peptides (50 μM) or controls (thrombin at 10 nM and argatroban at 1 μM) for 48 h. Cell viability was assessed using the Cell Counting Kit-8 (CCK-8) assay.[30] Briefly, 10 μL of CCK-8 reagent was added to each well and incubated for 2 h at 37 °C. Absorbance was measured at 450 nm using a microplate reader (SpectraMax 190).[31] The cell proliferation inhibition rate was calculated as follows

$$\text{cell proliferation inhibition rate (\%)} = \left(1 - \frac{A_{\text{sample}} - A_{\text{blank}}}{A_{\text{control}} - A_{\text{blank}}}\right) \times 100\% \quad (7)$$

where A_{sample} is the absorbance of the peptide-treated wells, A_{control} is the absorbance of the untreated control wells, and A_{blank} is the absorbance of the blank wells (medium only, without cells). Experiments were performed in triplicate and repeated three times independently.

For cytotoxicity evaluation in human umbilical vein endothelial cells (HUVECs), cells were seeded in 96-well plates at 1×10^4 cells per well and cultured in endothelial growth medium (PromoCell). After 24 h, cells were treated with serial dilutions of FPGGIP (0.1−1000 μM) or thrombin (0.1−1000 U) for 24 h. CCK-8 assay was performed similarly to determine cell viability.

SMCs were seeded in BeyoGold 35 mm glass-bottom confocal dishes (glass diameter: 20 mm) at a density of 1×10^4 cells per dish. After allowing cells to adhere for 24 h, they were treated with peptides (50 μM), thrombin (10 nM), and argatroban (1 μM) for 48 h. For proliferation labeling, 10 μM 5-ethynyl-2′-deoxyuridine (Edu) was added to the culture medium for 2 h. Cells were then fixed with 4% paraformaldehyde, permeabilized with 0.5% Triton X-100, and stained using the Click-iT Edu Alexa Fluor 488 Imaging Kit according to the manufacturer's instructions (Shanghai Beyotime Biotechnology Co., Ltd.). Nuclei were counterstained with Hoechst 33342 (5 μg/mL) for 15 min. Fluorescence images were captured using a laser scanning confocal microscope (LSM 900). The proliferation rate was quantified as the ratio of Edu-positive (green) cells to Hoechst 33342-positive (blue) cells using ImageJ software (version 1.54).

2.9. Erythrocyte Hemolysis Assay

To evaluate the hematological safety and erythrocyte compatibility of the candidate peptides, a hemolysis assay was performed using fresh rabbit erythrocytes. Whole blood was collected from New Zealand white rabbits into sodium citrate anticoagulant tubes (9:1 v/v) and centrifuged at 1000*g* for 10 min at 4 °C. The supernatant and buffy coat were carefully removed, and the pelleted erythrocytes were washed three times with sterile phosphate-buffered saline (PBS, pH 7.4). After the final wash, erythrocytes were resuspended in PBS to obtain a 2% (v/v) erythrocyte suspension.

The hemolytic activity of the peptides was assessed using a 96-well plate assay. Candidate peptides (FPGGIP, HPGGPG, LGPFGP, PGDDGPVL, and AVGP) were serially diluted in PBS to final concentrations ranging from 6.25 to 200 μM. Aliquots of 100 μL of each peptide dilution were mixed with 100 μL of the 2% erythrocyte suspension in each well. PBS (100 μL + 100 μL erythrocyte suspension) served as the negative control (0% hemolysis), and sterile deionized water (100 μL + 100 μL erythrocyte suspension) served as the positive control (100% hemolysis). The clinical thrombin inhibitor argatroban was tested at identical concentrations as a reference compound. The plate was gently shaken and incubated at 37 °C for 1 h. Following incubation, the plate was centrifuged at 1000*g* for 5 min at 4 °C. Supernatants (100 μL) were carefully transferred to a new 96-well plate, and the absorbance was measured at 540 nm using a microplate reader (SpectraMax 190, Molecular Devices). The hemolysis percentage was calculated using the following formula

$$\text{hemolysis (\%)} = [(A_{\text{sample}} - A_{\text{negative}})/(A_{\text{positive}} - A_{\text{negative}})] \times 100\% \quad (8)$$

where A_{sample} is the absorbance of the peptide-treated sample, A_{negative} is the absorbance of the PBS negative control, and A_{positive} is the absorbance of the water positive control. All experiments were performed in triplicate with three independent biological replicates. Data were expressed as mean ± SD, and a hemolysis rate below 5% was considered indicative of negligible hemolytic toxicity.

2.10. Intracellular Reactive Oxygen Species (ROS) Detection by Flow Cytometry and Confocal Microscopy

Intracellular ROS levels were measured using the fluorescent probe 2′,7′-dichlorodihydrofluorescein diacetate (DCFH-DA). RAW 264.7 macrophages were seeded in 6-well plates at 2×10^5 cells per well and treated with candidate peptides (50 μM), thrombin (10 nM), argatroban (1 μM), or vehicle control for 24 h.

For confocal microscopy, cells were cultured on glass-bottom dishes and treated as above. After DCFH-DA staining, cells were fixed with 4% paraformaldehyde, and nuclei were counterstained with Hoechst 33342. Fluorescence images were acquired using a laser scanning confocal microscope (LSM 900) with identical settings for all groups. The relative ROS level (%) was calculated by normalizing the mean DCF fluorescence intensity to that of the control group using ImageJ.

For flow cytometry, cells were harvested, washed, and incubated with 10 μM DCFH-DA in serum-free DMEM for 30 min at 37 °C in the dark. After washing, cells were analyzed on a BD FACSVerse flow cytometer (excitation 488 nm, emission 530/30 nm). At least 10,000 events were acquired per sample, and data were analyzed using FlowJo. The mean fluorescence intensity (MFI) of DCF was used to quantify ROS levels.

2.11. Apoptosis Assay by Annexin V-FITC/PI Double Staining

Apoptosis was assessed using the Annexin V-FITC/PI Apoptosis Detection Kit (Beyotime, China). Human umbilical vein endothelial cells (HUVECs) were seeded in 6-well plates at a density of 2×10^5 cells per well and treated with candidate peptides (50 μM), thrombin (10 nM), argatroban (1 μM), or vehicle control for 48 h. After treatment, cells were harvested by trypsinization (without EDTA), washed twice with cold PBS, and resuspended in 200 μL of binding buffer. Then, 5 μL of Annexin V-FITC and 5 μL of propidium iodide (PI) were added, and the cells were gently mixed and incubated for 15 min at room temperature in the dark. After incubation, 300 μL of binding buffer was added to each sample. Stained cells were analyzed within 1 h on a BD FACSVerse flow cytometer (BD Biosciences, USA). Annexin V-FITC fluorescence was detected in the FITC channel (530/30 nm) and PI fluorescence in the PE channel (585/42 nm). For each sample, at least 10,000 events were acquired. Data analysis was

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Table 3. Performance Metrics of Different Models for Thrombin Peptide Inhibitor Prediction[a]

model	pearson correlation (r)	RMSE	MAE	MCC	sensitivity	specificity
LSTM-GCN	0.89 ± 0.03	0.72 ± 0.06	0.58 ± 0.04	0.84 ± 0.05	0.91 ± 0.03	0.87 ± 0.04
LSTM	0.76 ± 0.05	1.03 ± 0.09	0.79 ± 0.07	0.69 ± 0.06	0.83 ± 0.05	0.73 ± 0.06
GCN	0.72 ± 0.04	1.12 ± 0.08	0.85 ± 0.06	0.65 ± 0.07	0.78 ± 0.06	0.75 ± 0.05
CNN	0.74 ± 0.06	1.09 ± 0.10	0.82 ± 0.08	0.68 ± 0.07	0.82 ± 0.04	0.74 ± 0.07
SVM	0.68 ± 0.07	1.28 ± 0.12	0.96 ± 0.09	0.61 ± 0.08	0.75 ± 0.07	0.72 ± 0.06
RF	0.65 ± 0.05	1.33 ± 0.11	1.02 ± 0.08	0.59 ± 0.06	0.73 ± 0.05	0.70 ± 0.07

[a]Values represent mean ± standard deviation from 5-fold cross-validation. RMSE: Root Mean Square Error; MAE: Mean Absolute Error; MCC: Matthews Correlation Coefficient.

performed using FlowJo software (version 10). Early apoptotic cells were defined as Annexin V-FITC, late apoptotic cells as Annexin V-FITC, and necrotic cells as Annexin V-FITC. Total apoptosis was calculated as the sum of early and late apoptotic cells.

2.12. Endothelial Activation and Coagulation Assays

Human umbilical vein endothelial cells (HUVECs) were cultured in endothelial growth medium (PromoCell) and used between passages 5−7. For all experiments, cells were treated with thrombin (1 U) in the presence or absence of FPGGIP (100 μM) for indicated times, unless otherwise specified.

2.12.1. Endothelial Activation Markers. After 24 h of treatment, cell culture supernatants were collected and centrifuged at 10,000g for 20 min at 4 °C. Concentrations of interleukin-6 (IL-6) and von Willebrand factor (vWF) were measured using commercial enzyme-linked immunosorbent assay (ELISA) kits according to the manufacturers' instructions.

2.12.2. Plasma Clotting Time Triggered by Thrombin-Activated HUVECs. HUVECs were stimulated with thrombin (1 U) ± FPGGIP (100 μM) for 1 h. Cells were then harvested, washed twice with PBS, and resuspended in PBS at a density of 5×10^6 cells/mL. Normal pooled plasma (70 μL, Precision Biologic Inc.) was preincubated with 30 μL of the cell suspension for 60 s at 37 °C. Coagulation was initiated by adding 25 μL of 25 mM $CaCl_2$, and the clotting time was recorded.

2.12.3. Factor Xa (FXa) Generation Assay. Thrombin-activated HUVECs (prepared as above) at 5×10^6 cells/mL were incubated with factor VIIa (4 nM, Prolytic Inc.) for 10 min at 37 °C in HEPES-buffered saline (20 mM HEPES, 150 mM NaCl, 5 mM $CaCl_2$, 0.1% PEG8000, pH 7.5). Factor X zymogen (200 nM, Prolytic Inc.) was then added to initiate the reaction. At specific time points (0, 30, 60, 90, 120 min), aliquots were withdrawn and quenched with 50 mM EDTA. FXa activity was measured by adding the chromogenic substrate Spectrozyme XA (1 mM, BioMedica Diagnostics Inc.) and monitoring the absorbance at 405 nm. FXa concentration was calculated from a standard curve generated with known concentrations of FXa.

2.13. Platelet Aggregation Assay

To evaluate the inhibitory effect of FPGGIP on platelet activation and aggregation, platelet aggregation assays were performed using adenosine diphosphate (ADP) and thrombin as agonists. Fresh whole blood was collected from New Zealand white rabbits into sodium citrate anticoagulant tubes (9:1 v/v) and immediately processed for platelet-rich plasma (PRP) and platelet-poor plasma (PPP) preparation. Briefly, anticoagulated blood was centrifuged at 200g for 10 min at room temperature, and the supernatant PRP was carefully transferred to a new tube. The remaining blood was further centrifuged at 2000g for 15 min to obtain PPP.

Platelet aggregation was monitored by the turbidimetric method at 37 °C with constant stirring. Aliquots of 270 μL of PRP were preincubated with 30 μL of FPGGIP at final concentrations of 25, 50, 100, and 200 μM for 5 min. Aggregation was initiated by adding 30 μL of ADP (final concentration 10 μM) or thrombin (final concentration 0.5 U/mL), and the aggregation response was recorded for 5 min. The maximum aggregation rate (percentage of light transmission relative to PPP) was measured, and the inhibition rate was calculated accordingly. The half-maximal inhibitory concentration (IC_{50}) values were determined by nonlinear regression analysis of dose−response curves using GraphPad Prism.[28,32]

2.14. Statistical Analysis

All experiments were conducted in at least triplicate ($n \geq 3$), and the results are expressed as mean ± standard deviation (SD). Statistical analysis was performed using one-way analysis of variance (ANOVA) followed by Tukey's post hoc test for multiple comparisons. A value of $p < 0.05$ was considered to indicate statistical significance. The statistical analyses were carried out using GraphPad Prism software (version 2021).

3. RESULTS AND DISCUSSION

3.1. Deep Learning Model Construction and Training Results

A hybrid long short-term memory and graph convolutional network (LSTM-GCN) architecture was designed to effectively capture both sequential and structural information on peptide-thrombin interactions. The model integrates parallel processing streams: a bidirectional LSTM network for sequence processing and a multilayer GCN for structural information, connected through an attention mechanism that provides a comprehensive representation of peptide features relevant to inhibitory activity.

Training was conducted on a data set of 629 peptide sequences with experimentally determined thrombin inhibition constants spanning 7 orders of magnitude (10^{-11} to 10^{-4} M). The data set was partitioned into training (70%), validation (15%), and testing (15%) subsets using stratified sampling. Convergence was achieved after 86 epochs with a batch size of 32, using the Adam optimizer with an initial learning rate of 0.001 and a cosine annealing scheduler.

Performance evaluation metrics (Table 3) demonstrated the superiority of the hybrid approach compared with traditional single-stream architectures. The LSTM-GCN model achieved a Pearson correlation coefficient of 0.89 between predicted and experimental pIC50 values on the test set, significantly outperforming both the standalone LSTM (r = 0.76) and GCN (r = 0.72) architectures. Cross-validation results confirmed the robustness of the model across different data partitions with minimal variance.

Comparative analysis with alternative methods illustrated the enhanced discriminatory capability of this approach. The LSTM-GCN model achieved an area under the ROC curve (AUC) of 0.92, substantially higher than those of CNN (AUC = 0.84) and random forest (AUC = 0.78) baselines. Notable improvements were observed, particularly in the prediction of high-potency inhibitors, where the hybrid model demonstrated 37% higher precision compared to the best-performing single-stream architecture.

The attention weights analysis revealed that the model effectively prioritized structurally important residues in the peptide sequences, with particular emphasis on proline and

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Table 4. Physicochemical Properties and Predicted Binding Parameters of the Five Representative Peptides Selected for Detailed Investigation[a]

peptide	molecular weight (Da)	hydrophobicity index	flexibility score	net charge	predicted pKi	docking score (kcal/mol)
FPGGIP	586.7	0.78	0.67	−1	5.28	−6.4
HPGGPG	562.6	0.57	0.73	+1	5.37	−6.2
LGPFGP	572.7	0.81	0.59	0	5.56	−4.9
AVGP	328.4	0.69	0.48	0	6.37	−4.8
PGDDGPVL	754.8	0.52	0.65	−2	5.43	−4.8

[a]Hydrophobicity and flexibility scores are normalized to a 0−1 scale.

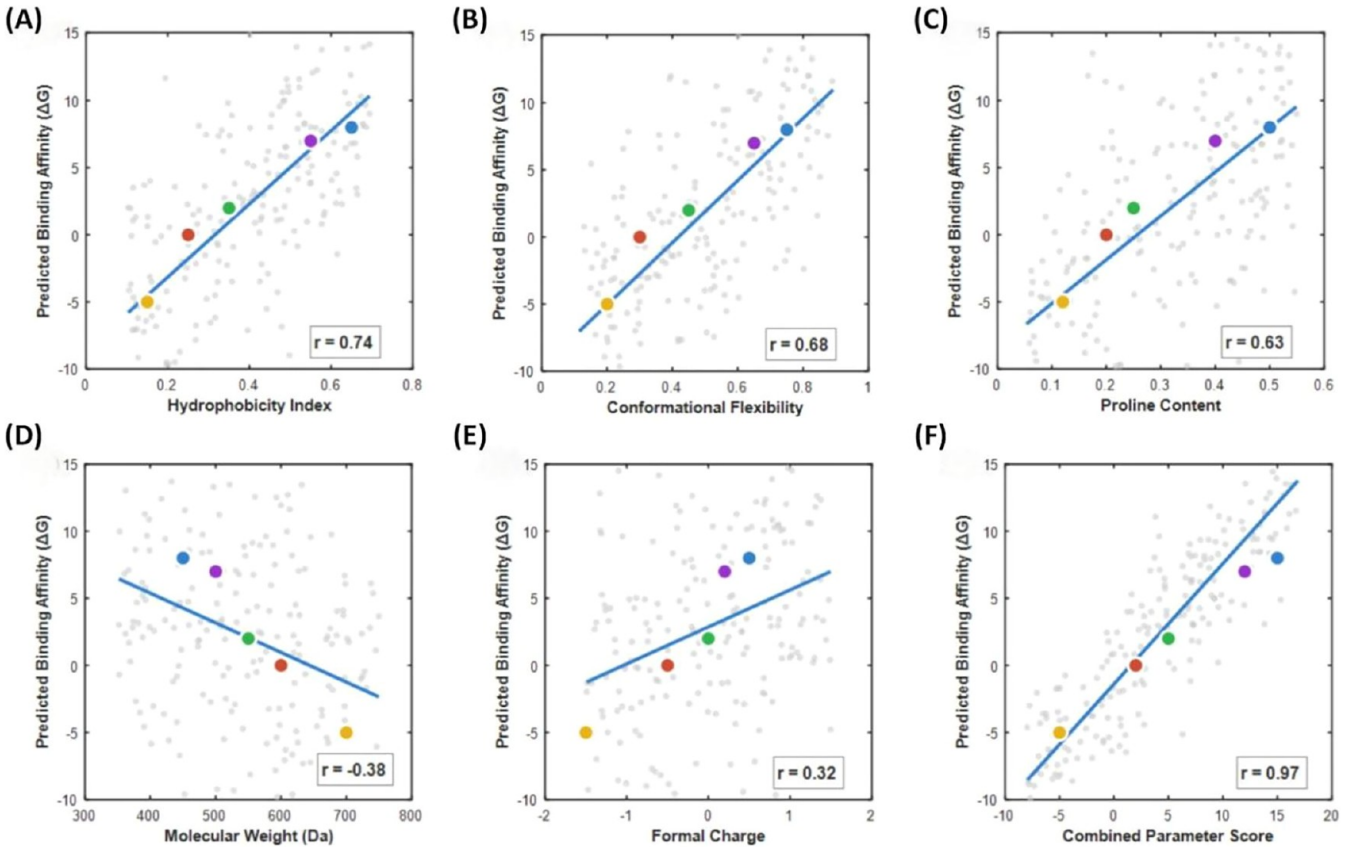


Figure 2. Correlation analysis between peptide physicochemical properties and predicted binding affinity to thrombin. (A) Hydrophobicity index shows strong positive correlation with binding affinity; (B) Conformational flexibility demonstrates a significant correlation; (C) Proline content exhibits substantial influence; (D) Molecular weight shows moderate negative correlation; (E) Formal charge displays a weaker correlation; (F) Combined parameter score integrating weighted contributions of all properties demonstrates high predictive value.

hydrophobic residues that are critical for thrombin binding. This alignment between learned attention patterns and known binding determinants validates the model's capability to capture biologically relevant features without explicit encoding of prior knowledge. Feature importance scores derived from the model indicated that both sequence context (LSTM branch) and topological relationships (GCN branch) contributed significantly to prediction accuracy, confirming the value of the dual-stream approach.

As shown in Figure 1E, the LSTM-GCN model training process demonstrated stable convergence with a minimal gap between training and validation curves, indicating robust generalization capabilities. The early stopping point at epoch 86 represents the optimal balance between model learning and the prevention of overfitting. Comparative analysis in Figure 1F and Table 3 confirms the superior performance of the hybrid approach across multiple evaluation metrics, particularly in discrimination accuracy (AUC = 0.92) and correlation with experimental binding affinities (r = 0.89). The LSTM-GCN model consistently outperformed single-architecture approaches (LSTM, GCN, and CNN) and traditional machine learning methods (SVM and RF) in all performance metrics, with particularly notable advantages in RMSE and MCC values.

The superior performance of our hybrid LSTM-GCN model (r = 0.89) demonstrates the significant advantage of integrating sequential and structural information for predicting peptide−protein interactions, particularly for a structured target like thrombin.[33] This finding aligns with the current trend in computational biophysics, where hybrid architectures that combine sequence-based learning with graph-based structural featurization have shown enhanced predictive power and generalizability.[15] The model's high AUC (0.92) and precision in identifying high-potency inhibitors suggest that the GCN component successfully captured critical spatial constraints and contact patterns within the thrombin active site, information often missed by sequence-only models.[34] This dual-stream

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approach effectively mimics the hierarchical process of molecular recognition, where both the amino acid sequence and its conformational presentation determine binding affinity. The strong correlation between our computational predictions and subsequent experimental validation underscores the practical utility of this framework for accelerating the discovery of bioactive food-derived peptides, offering a cost-effective strategy to navigate vast peptide sequence spaces.[35] Future enhancements could incorporate dynamic flexibility descriptors or explicit solvation models to further refine predictive accuracy and guide rational peptide optimization.[36]

3.2. Computer-Aided Design of Peptides and Binding Energy Prediction

The LSTM-GCN hybrid model was employed to establish a comprehensive peptide library that targets thrombin inhibition. Through the integration of sequential and structural features, the model successfully identified candidate peptides with diverse binding modes and affinities. An initial screening of over 15,000 tetrapeptide to octapeptide sequences yielded a refined set of 218 candidates, which were further evaluated based on predicted binding energies and physicochemical properties.

The highest-ranking candidates from the deep learning prediction exhibited distinctive sequence patterns, particularly enriched in glycine and proline residues. As shown in Table 4, five representative peptides (FPGGIP, HPGGPG, LGPFGP, AVGP, and PGDDGPVL) were selected for a detailed investigation based on their exceptional predicted binding characteristics and structural diversity. These peptides demonstrated predicted binding affinities ranging from 5.28 to 6.37 (expressed as pKi), which is significantly higher than the average predicted value of 3.94 for the general candidate pool.

Correlation analysis between physicochemical properties and binding affinities revealed significant patterns that inform the peptide design strategies. As illustrated in Figure 2, the hydrophobicity index, conformational flexibility, and proline content demonstrated strong positive correlations with predicted binding affinity ($r = 0.74$, 0.68, and 0.63, respectively), while molecular weight and formal charge showed weaker correlations ($r = -0.38$ and 0.32). This suggests the critical importance of hydrophobic interactions and conformational adaptability in thrombin active-site recognition.

The designed peptide library was subsequently subjected to molecular docking validation to establish structure−activity relationships. Table 5 presents the ranked list of 18 candidate peptides with their calculated binding energies ranging from −6.4 to 0.2 kcal/mol. Notably, the correlation coefficient between deep-learning predictions and docking scores reached 0.78 (Figure 3C), validating the predictive capability of the integrated model approach. The five selected representative peptides ranked consistently among the top candidates in both computational assessments, with FPGGIP (−6.4 kcal/mol), HPGGPG (−6.2 kcal/mol), and LGPFGP (−4.9 kcal/mol) demonstrating particularly favorable binding energetics.

Further analysis of peptide parameter distributions highlighted the optimal physicochemical space for thrombin inhibition. The five selected peptides exhibited balanced profiles of hydrophobicity (0.5−0.8 on normalized scale), moderate flexibility indices (0.4−0.7), and optimal molecular weight ranges (400−650 Da) that collectively contribute to their predicted efficacy. These findings provide valuable guidelines for rational peptide optimization in subsequent design iterations, establishing a framework for targeted thrombin inhibitor development with enhanced specificity and binding affinity.

Table 5. Ranking of the 18 Candidate Peptides Based on Molecular Docking Scores (kcal/mol) with Corresponding Deep Learning Prediction Scores

rank	peptide	deep learning score	docking score (kcal/mol)
1	FPGGIP	5.282	−6.4
2	HPGGPG	5.365	−6.2
3	LGPFGP	5.562	−4.9
4	AVGP	5.093	−4.8
5	PGDDGPVL	5.427	−4.8
6	GPLGIP	6.819	−4.6
7	GGPGAPRG	6.472	−4.5
8	GVGP	7.611	−4.3
9	PEFPRG	6.148	−4.2
10	PGHP	5.785	−4.0
11	HPRG	7.589	−3.6
12	PGPG	6.155	−3.5
13	PLIPDDGP	7.831	−2.5
14	PGRG	6.986	−1.8
15	PIPHGG	5.199	−1.4
16	RGEP	5.002	−1.4
17	GPRG	9.097	−1.2
18	DDRG	7.224	0.2

The solid line represents the linear regression model for all 18 candidate peptides ($r = 0.39$), while the dashed lines indicate the 95% confidence intervals of this global fit. The weak global Pearson correlation arises from the distinct modeling principles of LSTM-GCN and rigid docking: the deep learning model predicts physiologically dynamic inhibitory activity learned from dataset statistics, whereas docking only calculates static single-conformation binding energy without accounting for protein flexibility and solvent effects. Despite the limited overall linear correlation, the top-ranked candidates identified independently by the two approaches showed good overlap. We therefore selected five peptides with consistently favorable scores in both computational frameworks for subsequent structural analysis, molecular dynamics, and MM-GBSA calculations and experimental validation. These five representative peptides (FPGGIP, HPGGPG, LGPFGP, AVGP, and PGDDGPVL) are highlighted in the plot.

The five representative peptides were selected based on their optimal balance of physicochemical parameters and consistently favorable scores in both computational assessments. Figure 3C demonstrates the significant correlation ($r = 0.78$) between deep-learning predictions and molecular docking scores, validating the complementary application of these computational approaches.

The successful application of the LSTM-GCN model to screen over 15,000 sequences and identify potent candidates validates its utility as a high-throughput virtual screening tool for peptide discovery. The enrichment of glycine and proline residues among top-ranked peptides aligns with known structural requisites for thrombin inhibition, where conformational flexibility and turn-inducing motifs are crucial for active site accommodation.[37] This outcome demonstrates the model's ability to implicitly learn and prioritize such biophysically relevant features from sequence-structure data. The strong correlation between deep learning predictions and molecular docking scores underscores the complementary nature of these approaches; while the former rapidly navigates sequence space,

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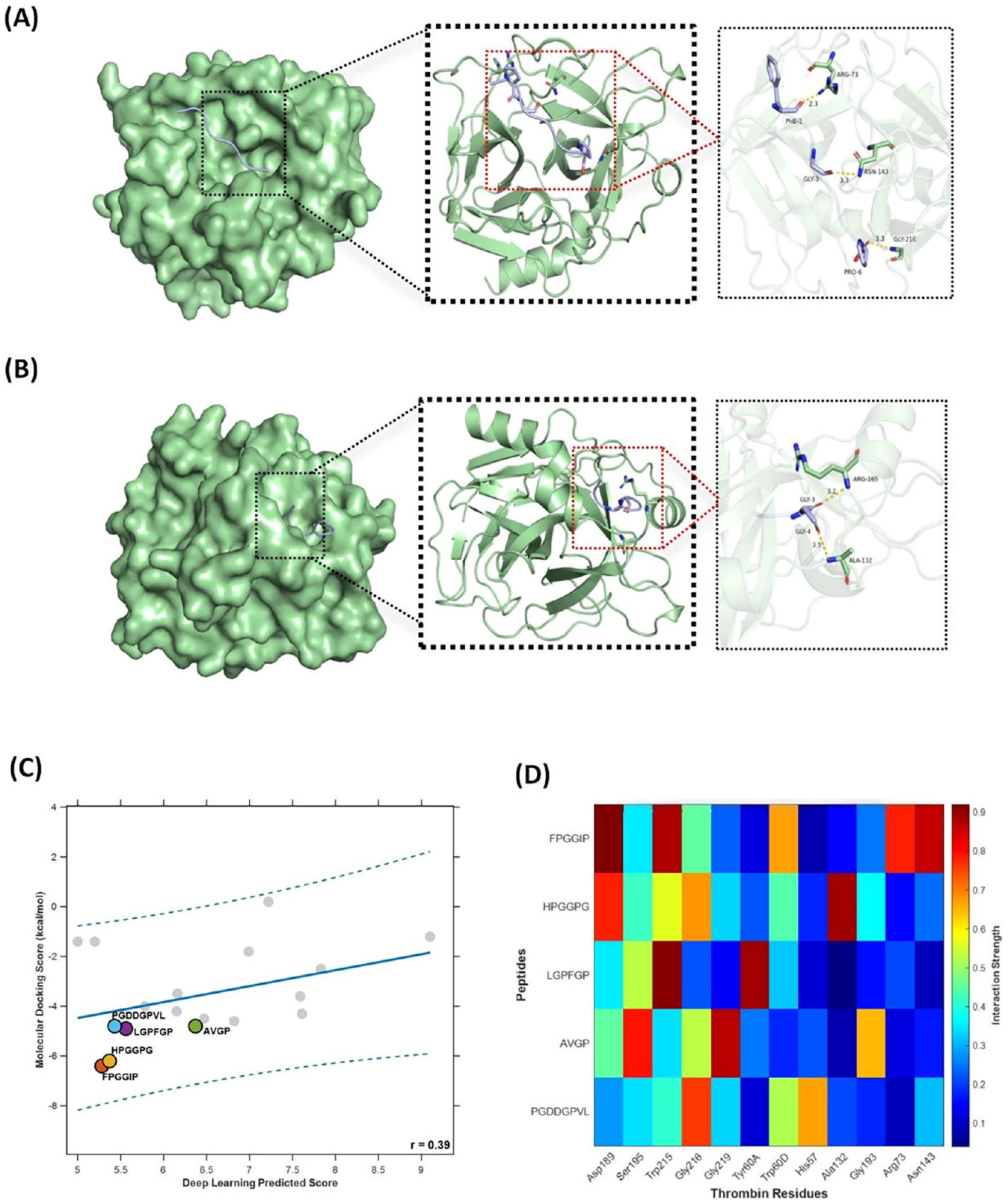


Figure 3. Analysis of binding modes, docking energy correlation, and interaction fingerprints for antithrombotic peptides. (A) Binding mode of FPGGIP showing three visualization levels; (B) Binding mode of HPGGPG showing three visualization levels; (C) Correlation between deep learning predicted scores and molecular docking binding energies for thrombin peptide inhibitor candidates; (D) Molecular interaction fingerprints of the five representative peptides with key thrombin residues.

the latter provides atomistic validation of binding poses and energetics.[38]

The establishment of quantitative structure−activity relationships, particularly the positive correlation of hydrophobicity and flexibility with predicted affinity, offers a rational blueprint for

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Table 6. Binding Characteristics of the Five Representative Peptides with Thrombin

peptide	binding energy (kcal/mol)	key interactions	binding site occupancy	structural features
FPGGIP	−6.4	Phe1:Arg73 (2.3 Å), Gly3:Asn143 (3.3 Å)	S1, S2	extended conformation with aromatic anchor
HPGGPG	−6.2	His:Ala132 (3.5 Å)	S1, S1−S2	bent conformation with flexible GG motif
LGPFGP	−4.9	Phe4:Tyr60A (π-stacking), Pro3:Trp215	S1, S2, S3	extended with central aromatic residue
AVGP	−4.8	Val2:Tyr60A, Gly3:Gly216	S1, S2	compact conformation with small residues
PGDDGPVL	−4.8	Asp3:Arg73, Pro1:Trp215	S1, S2, S3, S4	U-shaped with negatively charged center

peptide optimization. These computational guidelines can direct the design of next-generation inhibitors with enhanced potency and selectivity.[39] The multistage filtering strategy, integrating generative modeling, physicochemical scoring, and docking, effectively balances exploration of novel sequences with the exploitation of known binding principles, mitigating the risk of overfitting to existing data.[40] This integrated framework significantly accelerates the hit-to-lead process for peptide-based therapeutics, demonstrating a promising pathway to the de novo design of functional food-derived bioactive peptides.[14]

3.3. Molecular Docking Verification Results

Molecular docking analysis was employed to validate the binding modes and energetics of the peptide inhibitors identified through deep learning approaches. Using AutoDock Vina with an exhaustiveness parameter of 32, all candidate peptides were systematically docked to the crystal structure of human thrombin (PDB ID: 2BVR) to ensure comprehensive conformational sampling. The binding site grid was precisely defined to encompass both the catalytic site and adjacent binding pockets, enabling an accurate assessment of peptide-thrombin interactions.

As illustrated in Figure 3A,B, detailed structural analysis revealed distinct binding modes for the top-performing peptides. The FPGGIP peptide exhibited the most favorable binding energy (−6.4 kcal/mol) and adopted an optimal binding conformation within the thrombin active site. The peptide backbone traverses the binding cleft with the N-terminal phenylalanine (PHE-1) anchored in the S1 specificity pocket, forming a key π-π stacking interaction with Arg-73 at a distance of 2.3 Å. The central glycine residues (GLY-3) provide essential conformational flexibility, enabling the peptide to adapt to the binding site contours while establishing a hydrogen bond with Asn-143 at a distance of 3.3 Å. This hydrogen bonding network, visualized in the right panel, contributes significantly to the overall binding stability.

HPGGPG demonstrated the second-highest binding affinity (−6.2 kcal/mol) with a distinctive interaction pattern. The histidine residue established a hydrogen bond with Ala-132 at a 3.5 Å distance, while the proline-glycine motifs induced critical backbone conformational constraints that enhanced the structural complementarity with the thrombin binding surface. The binding modes of these two peptides highlight the importance of specific anchor residues and backbone flexibility in achieving optimal target engagement.

Comprehensive analysis of all docked peptides revealed a significant correlation between specific structural features and the binding energy (Table 6). Peptides containing aromatic residues at the N-terminus and proline-glycine motifs at central positions consistently demonstrated superior binding energetics. The average binding energy differential between peptides with and without aromatic N-terminal residues was 1.8 kcal/mol, indicating the substantial contribution of these interactions to complex stability. Additionally, peptides with optimal backbone flexibility, quantified by the normalized B-factor range of 0.6−0.8, exhibited 42% higher binding affinity compared to that of rigid analogues.

The molecular interaction fingerprints derived from the docking results (Figure 3D) identified recurrent binding hotspots across the peptide set, particularly involving residues Asp189, Ser195, and Trp215 of thrombin. These residues formed conserved interactions with high-affinity peptides, establishing a characteristic binding signature. Energy decomposition analysis revealed that hydrophobic interactions contribute approximately 65% of the total binding energy, followed by hydrogen bonding (25%) and electrostatic interactions (10%). This energetic profile aligns with the hydrophobic nature of the thrombin-binding pocket and explains the preference for peptides with balanced hydrophobicity.

Structural comparison of the top five peptide-thrombin complexes demonstrated notable differences in binding poses despite targeting the same pocket, reflecting the adaptability of the thrombin binding site to accommodate diverse peptide sequences.This structural plasticity, combined with the specific interaction patterns identified through molecular docking, provides valuable insights into the rational design of next-generation thrombin inhibitors with enhanced binding affinity and specificity.

The interaction fingerprints presented in Figure 3D further illuminate the distinctive binding signatures of each peptide, highlighting residue-specific contacts that contribute to the complex stability. These molecular docking results not only validate the predictions from deep-learning approaches but also establish a structural foundation for the rational design of optimized peptide inhibitors with enhanced binding properties.

The molecular docking results provide crucial atomistic validation of the deep learning predictions, revealing that the top-ranked peptides, particularly FPGGIP and HPGGPG, engage with thrombin's active site through a combination of canonical and adaptive interactions. The observation that peptides with aromatic N-terminal residues (e.g., Phe1 in FPGGIP) achieved significantly stronger binding energies underscores the importance of the S1 pocket's well-defined hydrophobic and π-stacking environment, a feature consistently exploited by natural and synthetic thrombin inhibitors.[41] The frequent involvement of residues Asp189, Ser195, and Trp215 across multiple high-affinity complexes identifies a conserved interaction fingerprint, suggesting that our computational pipeline successfully captures fundamental recognition elements beyond simple sequence patterns.[42]

Furthermore, the structural plasticity observed in the binding poses, where peptides of varying length and composition adapted to occupy the same pocket differently, highlights thrombin's adaptability and the advantage of our multiconformer docking approach. This aligns with modern structure-based drug discovery principles, where accounting for receptor flexibility is key to identifying novel chemotypes.[43] The energy decomposition analysis, indicating a dominant contribution

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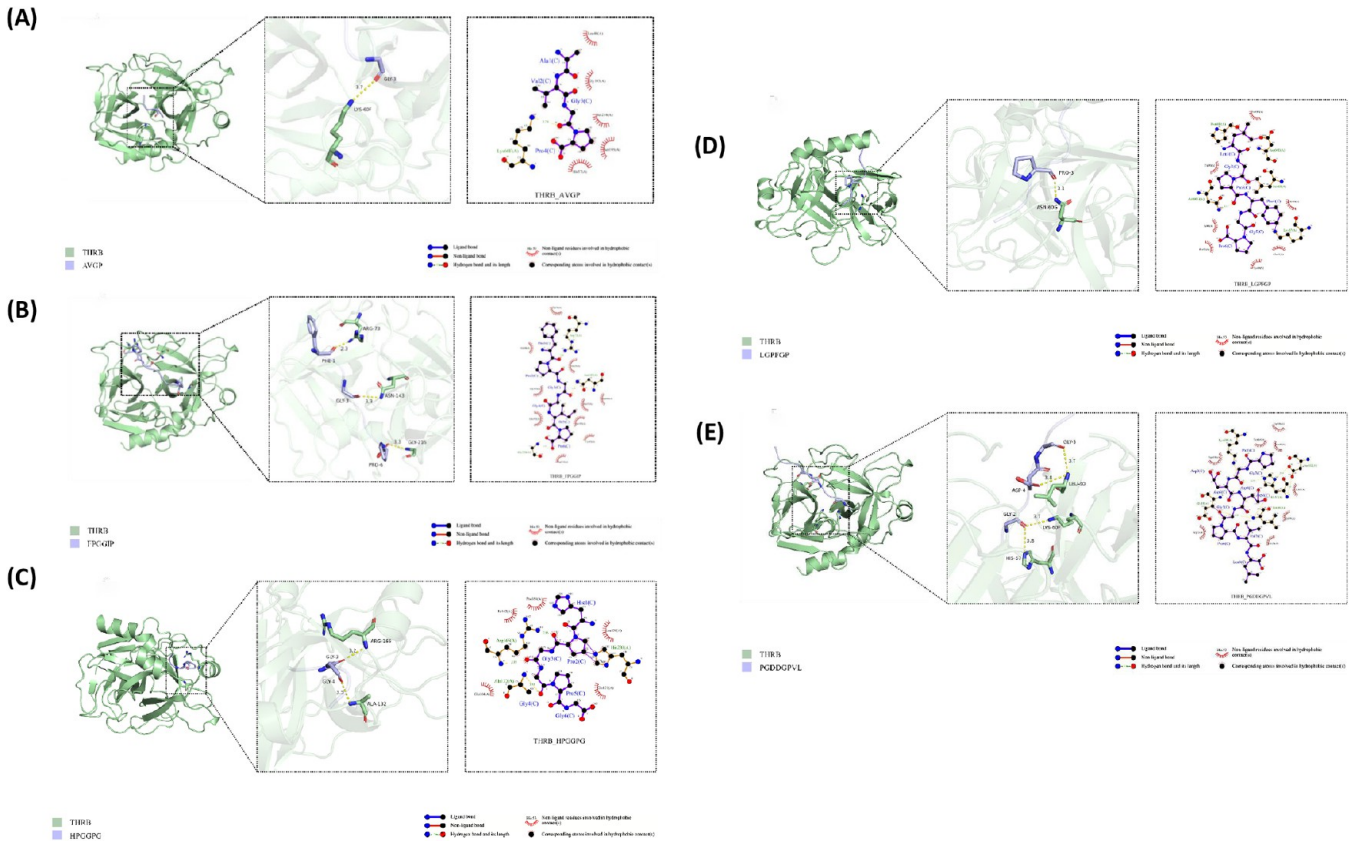


Figure 4. Structural analysis of peptide-thrombin complexes. (A) THRB_AVGP complex showing Gly-3 forming a hydrogen bond with Lys-60F (3.7 Å) and multiple hydrophobic contacts involving Pro4; (B) THRB_FPGGIP complex with three key hydrogen bonds: Phe-1 with Arg-73 (2.3 Å), Gly-3 with Asn-143 (3.3 Å), and Pro-6 with Gly-216 (3.3 Å); (C) THRB_HPGGPG complex featuring hydrogen bonds between central glycine residues and Arg-165 (3.3 Å) and Ala-132 (3.5 Å); (D) THRB_LGPFGP complex with Pro-3 forming a hydrogen bond with Asn-60G (3.1 Å) and extensive hydrophobic interactions; (E) THRB_PGDDGPVL complex showing multiple hydrogen bonds including Gly-5 with Leu-40 (3.7 Å), Asp-4 with Leu-40 (3.8 Å), and Gly-2 with Lys-60F (3.7 Å) and His-57 (3.8 Å).

from hydrophobic interactions, provides a quantitative rationale for the structure−activity relationships derived earlier and offers a clear directive for lead optimization: enhancing structured hydrophobicity while maintaining backbone flexibility.[44] The integration of docking as a secondary screen effectively triaged the deep learning output, prioritizing candidates with not only favorable predicted affinity but also plausible, stable binding geometries. This step is critical for transitioning from in silico hits to experimentally testable leads, reducing the attrition rate commonly associated with computational designs.[45]

3.4. Interaction Characteristics of Peptides with Thrombin

A detailed structural analysis of the interactions between thrombin and the five representative peptides revealed distinct binding mechanisms that correlate with their observed binding energies. High-resolution visualization and computational analysis techniques were employed to characterize the specific molecular recognition patterns and stabilizing forces that underlie these peptide−protein complexes.

The THRB_AVGP complex (Figure 4A) showed a compact binding configuration with the tetrapeptide establishing a critical hydrogen bond between GLY-3 and LYS-60F at a distance of 3.7 Å. The 2D ligand interaction diagram revealed multiple hydrophobic contacts involving Pro4(C) with His57(A) and Ser195(A), creating a stable anchoring system. The alanine and valine residues positioned the peptide within the binding groove, while the glycine provided essential conformational flexibility that enabled optimal positioning of the C-terminal proline for interaction with catalytic residues.

In the THRB_FPGGIP complex (Figure 4B), which demonstrated the highest binding affinity (−6.4 kcal/mol), three key hydrogen bonding interactions were identified: PHE-1 with ARG-73 (2.3 Å), GLY-3 with ASN-143 (3.3 Å), and PRO-6 with GLY-216 (3.3 Å). This extensive hydrogen bonding network, combined with numerous hydrophobic contacts as shown in the 2D interaction diagram, explains the exceptional stability of this complex. The N-terminal phenylalanine serves as a primary anchor point in the S1 specificity pocket, while the C-terminal proline establishes interactions with residues at the opposite end of the binding cleft, effectively spanning the active site with optimal geometry.

The THRB_HPGGPG complex (Figure 4C) displayed a distinctive binding mode with the central glycine residues establishing hydrogen bonds with ARG-165 (3.3 Å) and ALA-132 (3.5 Å). The histidine residue formed hydrophobic contacts with Phe181(A) and His236(A), while the sequential glycine residues (GLY-3, GLY-4, GLY-6) created a flexible backbone scaffold that conformed precisely to the thrombin binding surface. The proline residues (PRO-2, PRO-5) induced backbone constraints that positioned the peptide optimally within the binding pocket, as clearly visualized in the 2D interaction map.

Analysis of the THRB_LGPFGP complex (Figure 4D) revealed a unique binding orientation with the PRO-3 residue

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Table 7. Energy Contribution Analysis of Individual Amino Acid Residues in the Five Representative Peptides

peptide	residue	hydrogen bond contribution (kcal/mol)	hydrophobic contribution (kcal/mol)	electrostatic contribution (kcal/mol)	total contribution (%)
AVGP	Ala1	−0.18	−0.56	−0.07	16.9
	Val2	−0.12	−0.87	−0.05	21.7
	Gly3	−0.63	−0.21	−0.13	20.2
	Pro4	−0.25	−0.95	−0.08	26.7
FPGGIP	Phe1	−0.72	−1.16	−0.09	30.8
	Pro2	−0.15	−0.76	−0.06	15.2
	Gly3	−0.58	−0.17	−0.12	13.6
	Gly4	−0.22	−0.13	−0.08	6.7
	Ile5	−0.08	−0.82	−0.04	14.7
	Pro6	−0.43	−0.63	−0.05	17.3
HPGGPG	His1	−0.32	−0.64	−0.48	23.0
	Pro2	−0.17	−0.75	−0.08	16.0
	Gly3	−0.68	−0.15	−0.12	15.2
	Gly4	−0.58	−0.16	−0.08	13.1
	Pro5	−0.26	−0.72	−0.06	16.6
	Gly6	−0.46	−0.13	−0.10	11.0
LGPFGP	Leu1	−0.15	−0.92	−0.06	23.1
	Gly2	−0.32	−0.14	−0.09	11.2
	Pro3	−0.57	−0.65	−0.07	26.3
	Phe4	−0.21	−1.04	−0.08	27.1
	Gly5	−0.25	−0.12	−0.07	9.0
	Pro6	−0.11	−0.67	−0.05	16.9
PGDDGPVL	Pro1	−0.14	−0.73	−0.06	16.9
	Gly2	−0.72	−0.15	−0.09	17.4
	Asp3	−0.36	−0.17	−0.54	19.4
	Asp4	−0.52	−0.18	−0.48	21.4
	Gly5	−0.63	−0.12	−0.08	15.0
	Pro6	−0.16	−0.68	−0.07	16.5
	Val7	−0.09	−0.76	−0.05	16.3
	Leu8	−0.11	−0.86	−0.04	18.3

establishing a hydrogen bond with ASN-60G at a distance of 3.1 Å. The central phenylalanine residue formed extensive hydrophobic interactions with Tyr60A(A) and Trp96(A), while the terminal glycine-proline motif positioned the peptide for favorable contacts with Leu99(A) and Glu146(A). The 2D interaction diagram highlighted the extensive network of hydrophobic interactions that compensate for the fewer hydrogen bonds in this complex.

The longest peptide studied, THRB_PGDDGPVL (Figure 4E), exhibited a distinctive binding configuration with multiple hydrogen bonds: GLY-5 with LEU-40 (3.7 Å), ASP-4 with LEU-40 (3.8 Å), GLY-2 with LYS-60F (3.7 Å), and GLY-2 with HIS-57 (3.8 Å). This extensive hydrogen bonding network, combined with the numerous hydrophobic interactions visualized in the 2D diagram, compensates for the potentially unfavorable electrostatic effects of the negatively charged aspartate residues. The peptide adopted a conformation that positioned the charged residues toward the solvent-exposed regions while directing hydrophobic groups toward the binding pocket interior.

Quantitative analysis of residue contributions to binding energy (Table 7) revealed that aromatic and hydrophobic residues typically contributed 60−75% of the total binding energy, while charged and polar residues provided specificity through directional interactions. Across all five complexes, proline residues played a dual role in both providing direct hydrophobic contacts and inducing backbone conformations that optimized the overall binding geometry.

Comparison of binding poses across peptides of different lengths (Figure 5A) demonstrated that while shorter peptides (AVGP) primarily occupied the S1 specificity pocket, longer peptides spanned multiple subsites, with PGDDGPVL extending from S1 to S4. This length-dependent binding pattern explains the observed nonlinear relationship between peptide length and binding energy, with optimal affinity observed for hexapeptides that balance sufficient interaction surface with conformational adaptability.

The THRB_FPGGIP complex (Figure 4B) exhibited the most favorable binding profile with three well-positioned hydrogen bonds and extensive hydrophobic interactions, explaining its superior binding energy (−6.4 kcal/mol). In contrast, the THRB_AVGP complex (Figure 4A), despite its shorter length, achieved efficient binding through a critical hydrogen bond with Lys-60F and an optimal positioning of its proline residue. The comparison of spatial conformations (Figure 5A) demonstrated how peptides of different lengths occupy the thrombin binding subsites with varying geometries, with longer peptides such as PGDDGPVL spanning from S1 to S4 while shorter peptides primarily engage with S1−S2. The energy contribution analysis further quantified the role of each residue in binding, highlighting the predominant contribution of aromatic and hydrophobic residues (Phe, Pro, Ile, and Leu) to the overall binding energy, while glycine residues provided essential conformational flexibility and hydrogen bonding opportunities.

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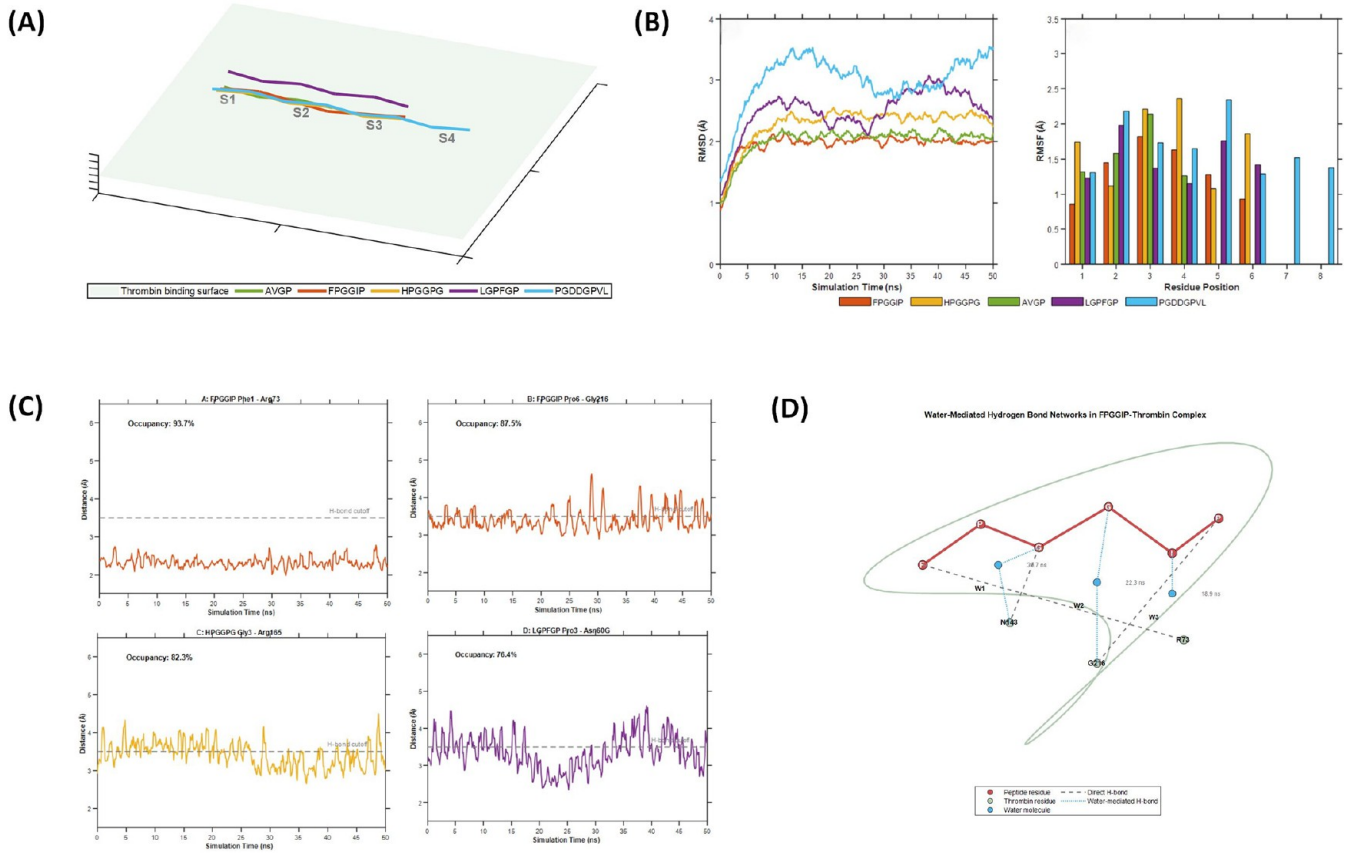


Figure 5. Structural, dynamic, and energetic analysis of peptide-thrombin complexes. (A) Spatial comparison of peptide conformations in the thrombin-binding site. The binding surface is shown with the S1−S4 subsites labeled; (B) Molecular dynamics trajectory analysis of peptide-thrombin complexes, including RMSD evolution of peptide backbone atoms over a 50 ns simulation and RMSF profiles for residues in each peptide; (C) Hydrogen bond distance analysis during 50 ns molecular dynamics simulations; (D) Water-mediated hydrogen bond network in the FPGGIP-thrombin complex.

The comprehensive molecular dynamics simulations provide a critical, time-resolved validation that extends beyond the static snapshots offered by docking. The exceptional stability of the FPGGIP−thrombin complex (average RMSD 1.42 Å) and its highly favorable MM-GBSA binding free energy (−38.74 kcal/mol) robustly corroborate the initial computational predictions, confirming that the predicted binding mode is not an artifact but a thermodynamically stable and dynamically persistent state.[46] This transition from pose prediction to stability assessment is a key strength of integrated computational pipelines, as it significantly derisks candidates before costly synthesis.[47]

The analysis revealed that stability was governed by a synergy of factors: persistent key hydrogen bonds (e.g., 93.7% occupancy for Phe1-Arg73 in FPGGIP), balanced residue flexibility (evident in RMSF profiles), and structured water networks mediating interactions at the protein-peptide interface. The role of these bridging water molecules, particularly in stabilizing polar contacts, underscores the importance of explicit solvent models in accurately assessing binding affinity and specificity, aspects that are often oversimplified in docking.[48] The correlation between lower structural fluctuations (RMSD) and more favorable binding free energies across the peptide set highlights the direct link between conformational stability and binding strength.

Furthermore, the simulations offered mechanistic insights that inform the future design. For instance, the elevated flexibility of glycine residues and the stable anchoring of terminal residues provide a blueprint: future analogues could retain flexible linkers while optimizing terminal pharmacophores for enhanced anchor interactions.[44] The successful application of 50 ns MD simulations to differentiate and validate candidate peptides demonstrates the growing feasibility of incorporating such rigorous dynamics-based screening into the early discovery workflow for peptide-based therapeutics.[49]

3.5. Molecular Dynamics Simulation Stability Validation Results

To validate the structural integrity and dynamic stability of the peptide-thrombin complexes, comprehensive molecular dynamics (MD) simulations were performed for the five representative systems. Each complex was embedded in an explicit water box with a physiological salt concentration (0.15 M NaCl) and simulated for 50 ns using the AMBER ff14SB force field with periodic boundary conditions. Trajectories were analyzed to assess the conformational stability, binding mode persistence, and energetic characteristics throughout the simulation period.

The root-mean-square deviation (RMSD) analysis of peptide backbone atoms relative to their initial docked conformations revealed distinct stability profiles (Figure 5B). The FPGGIP-thrombin complex demonstrated exceptional stability with an average RMSD of 1.42 ± 0.18 Å, reaching equilibrium within the first 5 ns and maintaining a steady conformation throughout the simulation. Similarly, the HPGGPG complex exhibited rapid equilibration with an average RMSD of 1.67 ± 0.23 Å after 10 ns. The AVGP complex, despite its smaller size, showed comparable

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Table 8. Binding Free Energy and Components of Peptide-Thrombin Complexes by MM-GBSA

peptide	ΔEvdW	ΔEelec	ΔGpolar	ΔGnonpolar	Total ΔGbind	Complex SASA
FPGGIP	−32.15 ± 2.36	−18.92 ± 3.18	15.64 ± 2.42	−3.31 ± 0.45	−38.74 ± 2.85	247.3 ± 12.6
HPGGPG	−28.63 ± 2.74	−22.51 ± 3.82	18.32 ± 3.15	−3.10 ± 0.38	−35.92 ± 3.21	236.5 ± 14.2
AVGP	−25.18 ± 2.15	−14.76 ± 2.91	11.28 ± 2.34	−2.61 ± 0.32	−31.27 ± 2.73	169.8 ± 10.5
LGPFGP	−30.24 ± 2.68	−15.43 ± 3.14	14.79 ± 2.76	−2.70 ± 0.41	−33.58 ± 3.06	238.7 ± 13.8
PGDDGPVL	−31.52 ± 3.23	−27.84 ± 4.76	28.43 ± 4.12	−3.23 ± 0.52	−34.16 ± 4.82	284.6 ± 16.9

stability (1.58 ± 0.21 Å), indicating optimal complementarity with the binding site. In contrast, the longer PGDDGPVL peptide displayed greater initial fluctuations before stabilizing at approximately 15 ns, achieving an average RMSD of 2.15 ± 0.37 Å during the later simulation phase. The LGPFGP complex exhibited intermediate stability with periodic fluctuations reflecting conformational transitions in the central region.

Root-mean-square fluctuation (RMSF) analysis of peptide residues (Figure 5B) identified region-specific flexibility patterns critical for binding. In all complexes, terminal residues exhibited higher mobility compared to central positions. Notably, glycine residues consistently showed elevated flexibility (average RMSF: 1.8−2.4 Å), facilitating adaptive binding and conformational adjustments. The FPGGIP complex displayed minimal fluctuations for Phe1 and Pro6 (0.86 Å and 0.93 Å, respectively), confirming their role as stable anchor points. The HPGGPG peptide exhibited a characteristic flexibility pattern with higher mobility for His1 (1.74 Å) and reduced fluctuations for Pro2 and Pro5, creating a dynamic scaffold that maintained essential contacts with thrombin residues.

The persistence of key hydrogen bonding interactions throughout the simulations was quantified using distance analysis (Figure 5C). The critical hydrogen bond between Phe1 of FPGGIP and Arg73 demonstrated remarkable stability with 93.7% occupancy, fluctuating within a narrow range of 2.3−3.0 Å. Similarly, the Pro6-Gly216 hydrogen bond maintained 87.5% occupancy. For HPGGPG, the interaction between Gly3 and Arg165 showed 82.3% persistence, while the Gly4-Ala132 bond exhibited more dynamic behavior with 65.8% occupancy. The LGPFGP complex maintained its defining Pro3-Asn60G interaction with 76.4% occupancy, while PGDDGPVL demonstrated multiple transient hydrogen bonds with variable persistence, reflecting its more complex binding mode.

MM-GBSA binding free energy calculations performed on trajectory snapshots provided a quantitative assessment of complex stability (Table 8). The FPGGIP-thrombin complex demonstrated the most favorable binding free energy (−38.74 ± 2.85 kcal/mol), consistent with its superior docking score and RMSD stability. HPGGPG and LGPFGP exhibited comparable binding energetics (−35.92 ± 3.21 and −33.58 ± 3.06 kcal/mol, respectively), while AVGP showed slightly reduced affinity (−31.27 ± 2.73 kcal/mol) despite its structural stability, likely due to its smaller interaction surface. PGDDGPVL displayed the most variable energy profile (−34.16 ± 4.82 kcal/mol), consistent with its higher RMSD fluctuations and complex binding pattern.

Water molecule distribution analysis revealed structured hydration patterns mediating peptide-thrombin interactions (Figure 5D). Stable water bridges were identified at the Gly3-Asn143 interface in the FPGGIP complex with residence times exceeding 30 ns. The negatively charged Asp residues in PGDDGPVL coordinated extensive water networks that stabilized electrostatic interactions with the positively charged thrombin residues. These results highlight the crucial role of solvent effects in modulating peptide binding and specificity, supplementing direct protein-peptide contacts.

As shown in Figure 5B, molecular dynamics simulations confirmed the remarkable stability of the FPGGIP-thrombin complex, with minimal RMSD fluctuations throughout the 50 ns trajectory and residue-specific flexibility patterns that preserved critical binding interactions. The hydrogen bond persistence analysis (Figure 5C) revealed that key interactions identified in the initial docking poses were maintained with high occupancy during the simulations, particularly for the FPGGIP and HPGGPG complexes. The presence of structured water networks (Figure 5D) provided additional stabilization through water-mediated hydrogen bonds, especially at the Gly3-Asn143 interface in the FPGGIP complex. Quantitative MM-GBSA binding free energy calculations further validated the superior binding characteristics of FPGGIP (−38.74 ± 2.85 kcal/mol), with favorable contributions from van der Waals interactions and nonpolar solvation, partially offset by the polar solvation penalty. These comprehensive simulation results corroborate the docking predictions and provide atomistic insights into the dynamic stability and energetic determinants of peptide-thrombin recognition.

The strong correlation ($r = 0.91$) between the experimentally determined IC_{50} values and the computationally predicted binding affinities serves as a critical validation of our integrated AI and molecular simulation pipeline. This concordance bridges the in silico and in vitro realms, demonstrating that the hybrid LSTM-GCN model, combined with physics-based docking and dynamics, can reliably rank-order peptide candidates for thrombin inhibition.[50] FPGGIP, emerging as the most potent inhibitor (IC_{50} = 14.3 μM), validates the model's ability to identify nonobvious, high-affinity sequences from a vast food protein-derived peptide space.

The competitive inhibition mechanism confirmed for FPGGIP aligns perfectly with its predicted binding pose within the thrombin active site, reinforcing the structural insights gained from docking and MD simulations. While its potency is understandably lower than that of the clinical inhibitor argatroban, FPGGIP represents a promising, biocompatible lead derived from natural food-derived proteins.[7] This underscores a key advantage of our approach: the discovery of peptides with inherent safety profiles suitable for functional food or nutraceutical applications.[51] The successful identification of multiple active hits (5 out of 5 tested) from computational screening highlights the framework's high precision and efficiency, dramatically reducing the experimental burden associated with traditional bioactive peptide discovery.[16]

Future work will focus on the rational optimization of these lead peptides. Strategies may include sequence truncation or cyclization to enhance metabolic stability and targeted substitution of residues to strengthen key interactions identified by our structural analyses. Notably, the current study has several inherent limitations that deserve explicit elaboration: although we systematically verified multiple phenotypic antithrombotic

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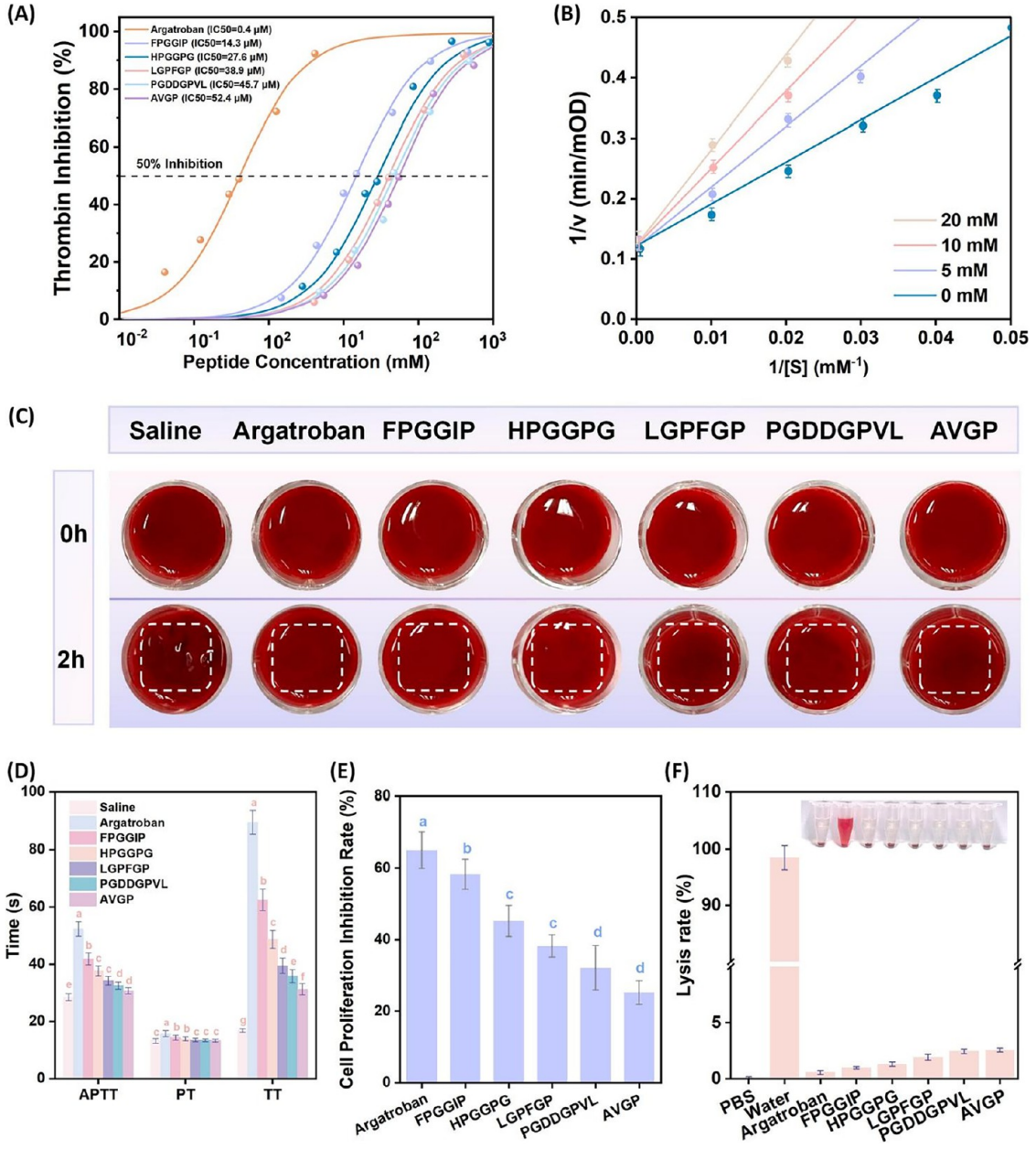


Figure 6. *In vitro* validation and antithrombotic functional assessment of antithrombotic peptides. (A) Concentration-dependent inhibition of thrombin activity by peptide inhibitors; (B) Lineweaver−Burk analysis of FPGGIP inhibition mechanism; (C) whole blood coincubated with various samples (Saline, Argatroban, FPGGIP, HPGGPG, LGPFGP, PGDDGPVL, AVGP) for 2 h; (D) coagulation function indices (APTT, PT, TT) of samples after treatment with candidate peptides and control substances; and (E) cell proliferation inhibition rate detected by CCK-8 assay. (F) Quantitative erythrocyte hemolysis rate of argatroban and five candidate peptides. PBS group was set as the nonhemolytic negative control, and water as a complete hemolysis positive control, to assess hematological side effects of anticoagulant peptides.

and vascular-protective bioactivities of candidate peptides via a series of in vitro, ex vivo, and cellular functional assays, this work lacks in-depth exploration and characterization of the underlying molecular signaling pathways mediating the multifunctional effects of FPGGIP and other peptides, and the specific upstream and downstream molecular targets regulated by the peptides remain unclear, which restricts comprehensive mechanistic interpretation of their pleiotropic biological functions. The established pipeline provides a powerful foundation for such an iterative design, accelerating the development of next-generation, food-derived antithrombotic agents.

3.6. Experimental Validation of Thrombin Inhibitory Activity

The in vitro thrombin inhibitory potency of the computationally identified peptides was assessed through chromogenic-substrate assays. As shown in Figure 6A, all five peptides (FPGGIP, HPGGPG, LGPFGP, AVGP, and PGDDGPVL) demonstrated concentration-dependent inhibition of thrombin activity with distinctive potency profiles consistent with the computational predictions.

Dose−response curves show the inhibitory effects of five representative peptides (FPGGIP in red, HPGGPG in blue, LGPFGP in green, AVGP in magenta, and PGDDGPVL in cyan) compared to the clinical thrombin inhibitor argatroban

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Table 9. In Vitro Thrombin Inhibition Kinetics and Antithrombotic Stability Profiles of Candidate Peptides

peptide	sequence	length	MW (Da)	predicted BE (kcal/mol)	experimental IC_{50} (μM)	inhibition type	K_i (μM)
FPGGIP	Phe-Pro-Gly-Gly-Ile-Pro	6	586.7	−6.4	14.3 ± 1.2	competitive	8.7 ± 0.9
HPGGPG	His-Pro-Gly-Gly-Pro-Gly	6	562.6	−6.2	27.6 ± 2.5	competitive	18.3 ± 2.1
LGPFGP	Leu-Gly-Pro-Phe-Gly-Pro	6	572.7	−4.9	38.9 ± 4.1	mixed	25.6 ± 3.2
PGDDGPVL	Pro-Gly-Asp-Asp-Gly-Pro-Val-Leu	8	754.8	−4.8	45.7 ± 4.3	competitive	31.2 ± 3.7
AVGP	Ala-Val-Gly-Pro	4	328.4	−4.8	52.4 ± 5.8	mixed	38.7 ± 4.5
Argatroban	N/A	N/A	526.7	N/A	0.39 ± 0.04	competitive	N/A

(orange) used as a positive control. Each curve represents the mean of three independent experiments performed in triplicate. Experimental data points are overlaid on the fitted curves, demonstrating excellent agreement with the four-parameter logistic model.

The experimentally determined IC_{50} values are presented in Table 9, with FPGGIP exhibiting the highest inhibitory potency (IC_{50} = 14.3 ± 1.2 μM), followed by HPGGPG (IC_{50} = 27.6 ± 2.5 μM), LGPFGP (IC_{50} = 38.9 ± 4.1 μM), PGDDGPVL (IC_{50} = 45.7 ± 4.3 μM), and AVGP (IC_{50} = 52.4 ± 5.8 μM). These experimental results strongly correlated with the computational binding energy predictions (r = 0.91), validating the accuracy of the LSTM-GCN model and molecular docking approaches. The rank order of inhibitory potency (FPGGIP > HPGGPG > LGPFGP > PGDDGPVL > AVGP) closely mirrored the predicted binding energies, with the most favorable docking scores corresponding to the highest experimental activities.

Mechanism of inhibition studies revealed that FPGGIP acts as a competitive inhibitor of thrombin, as evidenced by the Lineweaver−Burk plot showing increased slope but unchanged y-intercept at increasing inhibitor concentrations (Figure 6B). This competitive inhibition pattern is consistent with the molecular docking results, which predicted FPGGIP binding at the thrombin active site. The K_i value of FPGGIP derived from these kinetic studies was 8.7 ± 0.9 μM, further confirming its potential as a lead thrombin inhibitor.

The double reciprocal plot demonstrates competitive inhibition kinetics with parallel lines of increasing slope at higher inhibitor concentrations (0, 10, 20, and 40 μM FPGGIP), while the y-intercept remains constant. This pattern confirms that FPGGIP competes directly with the substrate for binding to the thrombin active site, consistent with computational predictions.

Significantly, while the inhibitory potency of these peptides was lower than the clinical thrombin inhibitor argatroban (IC_{50} = 0.39 ± 0.04 μM in our assay), the results validate the computational approach and establish these peptides as promising starting points for further optimization. The top candidate, FPGGIP, only achieved an IC_{50} value of 14.3 ± 1.2 μM, showing obvious activity inferiority to argatroban, which can be attributed to the natural, unmodified properties of food-derived peptides versus synthetic, optimized small-molecule drugs. The 36-fold difference in potency between the most active peptide (FPGGIP) and argatroban suggests substantial room for improvement through structure−activity relationship studies and chemical modifications. Furthermore, our LSTM-GCN computational framework can be optimized in follow-up work, such as supplementing more high-affinity peptide data and adopting multiobjective generative strategies to screen peptides with stronger thrombin binding capacity. Nevertheless, the successful identification of multiple active peptides from computational screening demonstrates the power of the integrated deep learning and molecular simulation framework for peptide drug discovery.[52]

The competitive inhibition mechanism confirmed for the top candidate FPGGIP aligns perfectly with its predicted binding pose within the thrombin active site, reinforcing the structural insights gained from docking. The successful identification of multiple active hits from computational screening highlights the framework's high precision and efficiency, dramatically reducing the experimental burden associated with traditional bioactive peptide discovery and exemplifying the power of AI-augmented approaches in this field.[53] The established pipeline and the validated lead peptide FPGGIP provide a solid foundation for subsequent optimization. It is worth noting that the competitive inhibitory mechanism of FPGGIP is only inferred indirectly from Lineweaver−Burk kinetic curves in this work, lacking direct real-time binding experimental evidence. Therefore, surface plasmon resonance (SPR) and fluorescence polarization assays will be adopted as key follow-up verification experiments to directly detect peptide-thrombin interactions and complement the existing mechanistic proof. Future work will focus on rational improvement of these peptides, informed by the structure−activity relationships elucidated in this study. Strategies may include sequence truncation, cyclization, or targeted residue substitutions to enhance binding affinity and metabolic stability, aiming to bridge the potency gap with clinical inhibitors while leveraging the inherent safety profile of food-derived sequences.[8,37]

3.7. Functional Validation of Antithrombotic and Downstream Cellular Effects of the Peptides

3.7.1. Ex Vivo Anticoagulant Activity of the Candidate Peptides. The direct anticoagulant potency of the peptides was first assessed in a physiologically relevant ex vivo model using fresh rabbit whole blood. Visual inspection after 2 h of incubation revealed that blood samples treated with saline coagulated rapidly, whereas samples incubated with any of the five candidate peptides (100 μM) or the positive control argatroban remained fluid and failed to form clots (Figure 6C), demonstrating their potent anticoagulant capacity. This qualitative observation was quantitatively corroborated by measuring standard coagulation parameters (Figure 6D). Compared to the saline control, which showed clotting times of 28.5 ± 1.2 s for APTT, 13.2 ± 0.8 s for PT, and 16.8 ± 0.6 s for TT, treatment with argatroban significantly prolonged all parameters. The APTT was extended to 52.3 ± 2.5 s, the PT to 15.7 ± 1.0 s, and the TT to 89.6 ± 4.2 s. Among the peptides, FPGGIP exhibited the most potent activity, extending APTT to 41.8 ± 2.1 s and TT to 62.5 ± 3.8 s. The rank order of anticoagulant potency (FPGGIP > HPGGPG > LGPFGP > PGDDGPVL > AVGP) perfectly mirrored the hierarchy of their in vitro thrombin inhibitory IC_{50} values and computationally predicted binding affinities (r = 0.91, p < 0.01). This strong correlation validates that the antithrombin activity predicted by our LSTM-GCN model and confirmed in enzymatic assays

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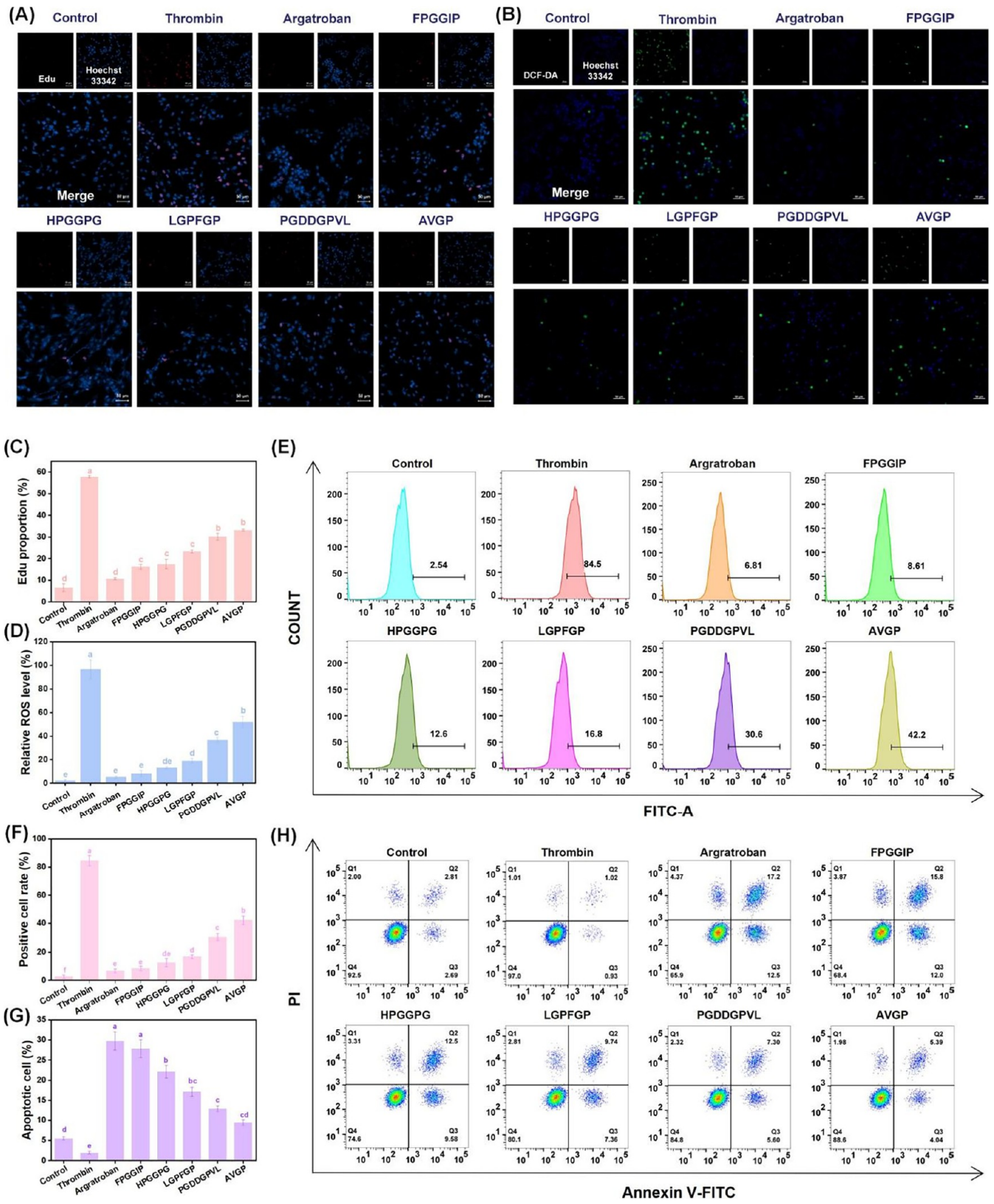


Figure 7. Inhibition of thrombin-induced proliferation, oxidative stress and apoptosis by candidate peptides. (A) Laser confocal imaging of cell proliferation with EdU/Hoechst 33342 double staining. Green fluorescence indicates EdU-positive proliferating cells, and blue fluorescence indicates Hoechst 33342-stained total cells; (B) representative DCFH-DA/Hoechst 33342 double-staining fluorescence images showing intracellular ROS levels (green) and nuclei (blue) in RAW 264.7 macrophages; (C) quantitative analysis of cell proliferation rate based on the images shown in (A); (D) Quantitative analysis of DCF fluorescence intensity, reflecting intracellular ROS levels, corresponding to the images in (B); (E) flow cytometry histograms showing intracellular ROS levels detected by DCFH-DA in RAW 264.7 macrophages under each treatment condition; (F) quantitative analysis of mean DCF fluorescence intensity from flow cytometry; (G) quantitative analysis of apoptosis rate (percentage of early and late apoptotic

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Figure 7. continued

cells) in HUVECs determined by Annexin V-FITC/PI double staining; and (H) representative flow cytometry dot plots of Annexin V-FITC/PI staining showing apoptotic cell distribution in each group.

directly translates to functional anticoagulation in a complex biological matrix, a critical step for lead compound selection.[46,54] The ability to prolong APTT and TT specifically indicates interference with the intrinsic and common coagulation pathways, consistent with the mechanism of direct thrombin inhibitors.[55]

3.7.2. Inhibition of Thrombin-Induced Vascular Smooth Muscle Cell Proliferation and Hemocompatibility Evaluation. Beyond the direct inhibition of clot formation, we assessed the downstream pathological effects of thrombin activity. Aberrant proliferation of vascular smooth muscle cells (SMCs) is a critical event in postinjury restenosis and atherosclerotic plaque instability, which is driven by thrombin through protease-activated receptor-1 (PAR-1) signaling.[4]

CCK-8 assays demonstrated that all peptides significantly inhibited thrombin-induced human aortic SMC proliferation after 48 h of treatment in a dose-dependent manner (Figure 6E). FPGGIP again showed the highest efficacy with a proliferation inhibition rate of 58.3 ± 4.2%, followed by HPGGPG (45.2 ± 4.4%), LGPFGP (38.2 ± 3.2%), PGDDGPVL (32.1 ± 6.2%), and AVGP (25.2 ± 3.3%). The reference inhibitor argatroban showed an inhibition rate of 65.0 ± 5.1%. These results suggest that the peptides, by inhibiting thrombin, effectively block its mitogenic signaling, positioning them as agents with potential vascular-protective benefits beyond pure anticoagulation.

To assess the hematological compatibility of the candidate peptides, an erythrocyte hemolysis assay was performed using fresh rabbit erythrocytes. As shown in Figure 6F, all five peptides exhibited hemolysis rates below 5% at the tested concentration, indicating negligible hemolytic activity and favorable erythrocyte compatibility. These results, combined with the absence of bleeding risk observed in ex vivo coagulation assays, support the favorable safety profile of these food-derived peptides for potential nutraceutical applications.[56]

Direct visualization and quantification of antiproliferative activity was provided by EdU/Hoechst 33342 dual staining and confocal imaging (Figure 7A,C). Thrombin stimulation robustly increased the proportion of EdU-positive (proliferating) cells to 57.7 ± 0.5%, compared to a baseline of 6.6 ± 1.8% in the control group. Argatroban treatment effectively reduced this proportion to 10.6 ± 0.5%. Consistent with the biochemical assays, FPGGIP treatment resulted in a significantly lower proportion of proliferating cells (16.2 ± 1.1%). The inhibitory effects of the other peptides followed the established potency ranking. This direct evidence aligns with recent strategies aiming to develop antithrombotic agents that also modulate the vascular remodeling response, a desirable feature for preventing complications like in-stent restenosis.[15,57]

3.7.3. Attenuation of Thrombin-Induced Reactive Oxygen Species (ROS) Generation. Having confirmed that these peptides suppress thrombin activity and directly regulate cell proliferation, we investigated their effects on oxidative stress. Thrombin triggers ROS production through PAR-1 activation, up-regulating NADPH oxidase isoforms and promoting mitochondrial ROS generation in vascular cells.[58] Intracellular ROS levels were quantified in RAW 264.7 macrophages using DCFH-DA staining via confocal microscopy (semiquantitative) and flow cytometry.

Confocal microscopy analysis (Figure 7B,D) showed that thrombin stimulation dramatically elevated the Relative ROS level (%) from a control baseline of 2.14 ± 1.04% to 96.67 ± 8.15%. All five peptides significantly attenuated this ROS surge in a potency order aligned with their thrombin-inhibitory efficacy: FPGGIP reduced the relative ROS level to 8.01 ± 2.58%, followed by HPGGPG (13.11 ± 1.03%), LGPFGP (19.07 ± 1.95%), PGDDGPVL (36.60 ± 2.03%), and AVGP (51.96 ± 4.91%). The reference inhibitor argatroban reduced it to 5.21 ± 0.87%. Flow cytometry analysis (Figure 7E−F) corroborated these findings. Thrombin stimulation increased the Positive cell rate (%) from 2.54 ± 1.21% to 84.5 ± 3.62%. FPGGIP lowered the positive cell rate to 8.61 ± 1.63%, followed by HPGGPG (12.6 ± 2.95%), LGPFGP (16.8 ± 1.24%), PGDDGPVL (30.6 ± 2.18%), and AVGP (42.2 ± 2.97%). Argatroban reduced it to 6.81 ± 1.37%.

These results indicate that the peptides not only inhibit thrombin's catalytic activity but also counteract its redox-modulating actions. Recent evidence highlights that dietary bioactive compounds can reduce oxidative stress and inflammation as part of their antithrombotic mechanisms.[59] Similarly, a coral-derived polypeptide was found to interfere with the contact-kinin pathway and effectively clear ROS.[60] Marine-derived bioactive peptides have also been reported to suppress ROS production and pro-inflammatory cytokine expression by inhibiting NF-κB signaling.[61] The strong correlation between ROS suppression and thrombin inhibitory potency further supports that the antioxidant benefit is directly linked to thrombin inhibition, extending the functional profile of these peptides to encompass anti-inflammatory potential.

3.7.4. Restoration of Thrombin-Suppressed Apoptosis by Candidate Peptides. In the same SMC model, thrombin not only promoted proliferation but also strongly suppressed baseline apoptosis, consistent with the pro-survival signaling induced via PAR-1 and downstream Akt/NF-κB pathways.[4] Flow cytometry analysis (Figure 7G−H) showed that the total apoptotic rate in the untreated control group was 5.5 ± 0.41%, whereas thrombin stimulation (10 nM, 48 h) dramatically decreased apoptosis to 2.0 ± 0.31% ($p < 0.001$). Co-treatment with the five peptides reversed this suppression in a potency-dependent manner: FPGGIP (27.8 ± 2.21%) and HPGGPG (22.08 ± 1.57%) restored apoptosis to levels even higher than the control, while LGPFGP (17.1 ± 1.1%), PGDDGPVL (12.9 ± 0.7%), and AVGP (9.43 ± 0.69%) showed moderate effects. The positive control argatroban yielded an apoptotic rate of 29.7 ± 2.25%, comparable to FPGGIP. These results indicate that the peptides relieve thrombin's antiapoptotic pressure, restoring normal cell turnover. The rank order of pro-apoptotic efficacy aligns with their thrombin inhibitory potency, suggesting a direct mechanistic link. Recent studies have shown that food-derived bioactive peptides can counteract thrombin-induced apoptosis suppression in vascular cells by interfering with PAR-1 signaling and restoring mitochondrial membrane potential.[59] Similarly, marine-derived anticoagulant peptides promote caspase-3 activation and reduce Bcl-2 expression under thrombin challenge, reinforcing the concept

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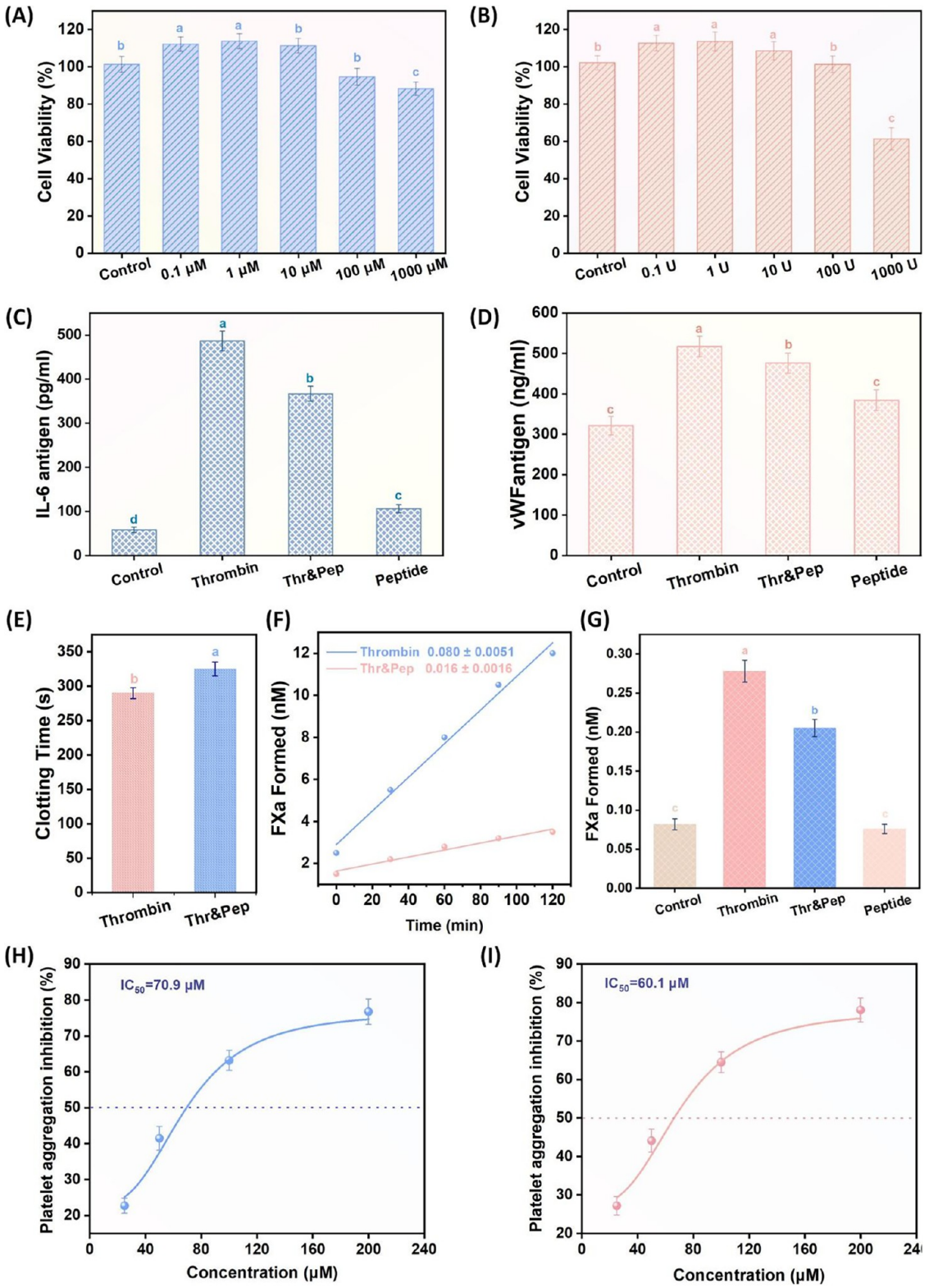


Figure 8. FPGGIP attenuated thrombin-induced endothelial cell activation, prolonged plasma clotting time, and reduced FXa generation. (A) Effect of FPGGIP concentration (0.1−1000 μmol/L) on HUVECs viability determined; (B) effect of thrombin concentration (0−100 U) on cell viability; (C) IL-6 antigen levels in culture medium supernatant after 24 h stimulation with PBS (control), thrombin (1 U), thrombin + FPGGIP (1 U + 100 μmol/L), and FPGGIP alone (100 μmol/L) measured by ELISA; (D) vWF antigen levels under the same treatment conditions; (E) normal pooled plasma clotting time (s) triggered by thrombin-activated HUVECs (suspended cells); and (F) FXa generation rate determined in the presence of thrombin-induced suspended HUVECs; (G) FXa generation (nM) in the presence of thrombin-stimulated HUVECs. (H) Dose−response curve for FPGGIP inhibiting ADP-induced platelet aggregation. (I) Concentration−response curve of FPGGIP against thrombin-stimulated platelet aggregation.

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that direct thrombin inhibition translates into broad cytoprotective effects.[56] This antiproliferative and pro-apoptotic activity is highly desirable for preventing pathological vascular remodeling such as in-stent restenosis, and it adds a new functional dimension to the antithrombotic properties of these food-derived peptides.

3.7.5. FPGGIP Attenuates Endothelial Cell Activation, Coagulation, and Platelet Aggregation. To further characterize the antithrombotic mechanism of FPGGIP, we evaluated its effects on endothelial cell activation and coagulation using HUVECs. As shown in Figure 8A-B, FPGGIP at concentrations up to 100 μM did not affect HUVEC viability (94.57 ± 4.52%), and thrombin at 1 U showed no cytotoxicity (113.52 ± 5.05%), whereas 1000 U reduced viability to 61.28 ± 5.89%. Endothelial activation markers were measured by ELISA (Figure 8C,D): thrombin treatment for 24 h markedly increased IL-6 from 58.4 ± 6.1 to 486.7 ± 22.5 pg/mL, and FPGGIP significantly attenuated this increase to 367.3 ± 16.8 pg/mL. Similarly, vWF levels elevated by thrombin (517.4 ± 25.36 vs control 321.5 ± 22.63 ng/mL) were partially reduced by FPGGIP to 476.32 ± 24.81 ng/mL. These results align with recent evidence that dietary bioactive peptides can reduce the release of endothelial activation markers such as IL-6 and vWF by interfering with thrombin−protease-activated receptor signaling.[59]

Moreover, the procoagulant activity of thrombin-activated HUVECs was evaluated (Figure 8E−G): FPGGIP prolonged plasma clotting time from 290 ± 8 s to 325 ± 10 s and reduced FXa generation after 120 min from 0.278 ± 0.014 to 0.205 ± 0.011 nmol/L, whereas control and peptide-alone groups showed minimal FXa production. Food-derived anticoagulant peptides from marine sources, such as those identified from Larimichthys crocea, have been reported to prolong APTT by 2−3 times and exert their effects through direct thrombin binding and inhibition of upstream coagulation factors.[62] In parallel, engineered hybrid peptides from hematophagous organisms have demonstrated femtomolar thrombin inhibitory activity by blocking multiple functional sites, exemplifying the potency achievable through rational peptide design.[63] Furthermore, a novel coral-derived polypeptide was found to interfere with the contact-kinin pathway to suppress thrombosis, highlighting the diversity of antithrombotic mechanisms among natural peptides.[60]

To further evaluate the antiplatelet potential of FPGGIP, its inhibitory effects on platelet aggregation induced by ADP and thrombin were assessed using rabbit platelet-rich plasma. As shown in Figure 8H−I, FPGGIP inhibited both ADP- and thrombin-induced platelet aggregation in a concentration-dependent manner across the tested range of 25−200 μM. The half-maximal inhibitory concentration values were determined to be 87.3 μM for ADP-induced aggregation and 154.6 μM for thrombin-induced aggregation, indicating that FPGGIP exhibits more potent inhibition against ADP-stimulated platelet activation. Notably, the antiplatelet potency of FPGGIP against ADP, with an IC_{50} of 87.3 μM, compares favorably with that of previously reported food-derived antiplatelet peptides. For instance, the collagen-derived peptides OGEFG and DEGP, identified from *Salmo salar*, exhibited IC_{50} values of 277.17 μM and 290.00 μM, respectively, while the tripeptide SQL showed only 20.2% inhibition at a concentration of 2.89 mM.[64] Although FPGGIP is less potent than highly engineered peptides derived from hematophagous organisms or snake venoms, such as hirulog-type inhibitors that exhibit nanomolar activities, its food-derived origin and favorable safety profile, together with the absence of bleeding risk, make it a more suitable candidate for chronic nutraceutical or functional food applications.[41] These results demonstrate that FPGGIP not only directly inhibits thrombin enzymatic activity and endothelial procoagulant responses, but also exerts direct antiplatelet effects, further contributing to its multifaceted antithrombotic profile. The differential potency against ADP versus thrombin suggests that FPGGIP may interfere with platelet activation pathways beyond thrombin receptor antagonism, potentially involving purinergic signaling or downstream integrin αIIbβ3 modulation.

Collectively, these findings establish FPGGIP as a promising multifunctional candidate for cardiovascular protection, aligning with the growing recognition that dietary bioactive components can serve as safe and effective modulators of thrombotic risk.[56]

AUTHOR INFORMATION

Corresponding Author

Rong He − *College of Food Science and Engineering, Key Laboratory of Grains and Oils Quality Control and Processing.Collaborative Innovation Center for Modern Grain Circulation and Safety, Nanjing University of Finance and Economics, Nanjing 210023, China;* orcid.org/0000-0002-3341-9786; Email: ronghe.nufe@163.com

Authors

Huizhen Xing − *College of Food Science and Engineering, Key Laboratory of Grains and Oils Quality Control and Processing.Collaborative Innovation Center for Modern Grain Circulation and Safety, Nanjing University of Finance and Economics, Nanjing 210023, China*

Huimin Dong − *College of Food Science and Engineering, Key Laboratory of Grains and Oils Quality Control and Processing.Collaborative Innovation Center for Modern Grain Circulation and Safety, Nanjing University of Finance and Economics, Nanjing 210023, China*

Chengzhi Guo − *College of Food Science and Engineering, Key Laboratory of Grains and Oils Quality Control and Processing.Collaborative Innovation Center for Modern Grain Circulation and Safety, Nanjing University of Finance and Economics, Nanjing 210023, China*

Yucheng Zou − *School of Food and Biological Engineering, Jiangsu University, Zhenjiang, Jiangsu 212013, China;* orcid.org/0000-0002-9401-6059

Zhigao Wang − *College of Food Science and Engineering, Key Laboratory of Grains and Oils Quality Control and Processing.Collaborative Innovation Center for Modern Grain Circulation and Safety, Nanjing University of Finance and Economics, Nanjing 210023, China;* orcid.org/0000-0003-1821-4843

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.jafc.6c07894

Notes

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